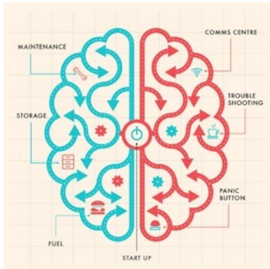
Object recognition
in ventral visual cortex and deep networks



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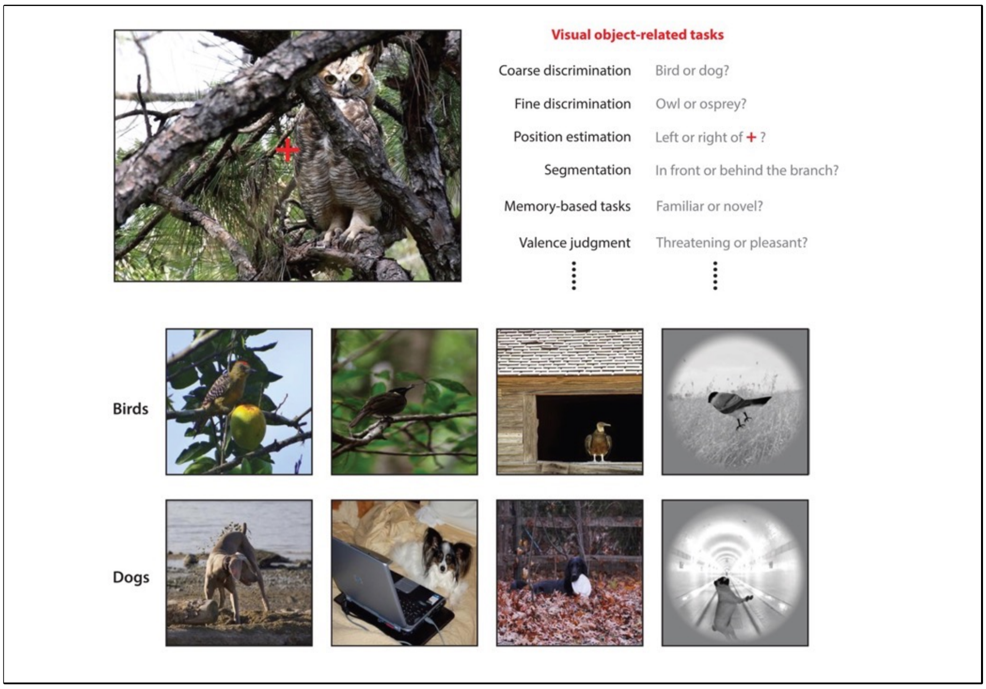
Cognition and Neuroscience
Second cycle Degree in Artificial Intelligence – 2024/25



Visual object-related tasks
Coarse discrimination
Bird or dog?
Fine discrimination
Owl or osprey?
Position estimation
Left or right of + ?
Segmentation
In front or behind the branch?
Memory-based tasks
Familiar or novel?
Valence judgment
Threatening or pleasant?
Birds
Dogs

What is vision?

What does it mean, to see?
Vision is the process of discovering what is present in the world, and where it is. (Marr, Vision, 1982)

Vision is a process that produces from images of the external world a description that is useful to the viewer and not cluttered with irrelevant information (Marr and Nishihara, 1978).

Vision dominates our perceptions and memories of the world and appears even to frame the way we think.

Vision is used not only for object recognition but also for guiding our movements.
These separate functions are mediated by at least two parallel and interacting pathways.

Vision, and more generally the brain, is a system that analyzes information (information processing device): receives inputs and transforms them into outputs.

What is vision?

“What does it mean, to see? The plain man's answer (and Aristotle’s, too) would be, to know what is where, by looking. In other words, vision is the process of discovering what is present in the world, and where it is.
Vision is therefore, first and foremost, an information-processing task, but we cannot think of it just as a process.
For if we are capable of knowing what is where in the world, our brains must somehow be capable of representing this information.
The study of vision must therefore include not only the study of how to extract from images the various aspects of the world that are useful to us, but also an inquiry into the nature of the internal representations by which we capture this information and thus make it available as a basis for decisions about our thoughts and actions.
This duality the representation and the processing of information-lies at the heart of most information-processing tasks and will profoundly shape our investigation of the particular problem posed by vision”

(David Marr, Vision, 1982)

Marr’s “tri-level hypothesis”

“arr argued that to truly understand vision — biological or artificial — we have to approach it on multiple levels: computational, algorithmic, and implementational. That framework still guides how we think about perception today, whether we’re studying the human brain or training a neural net

1.2 Understanding Complex Information-Processing Systems

Computational theory	Representation and algorithm	Hardware implementation
What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?	How can this computational theory be implemented? In particular, what is the representation for the input and output, and what is the algorithm for the transformation?	How can the representation and algorithm be realized physically?

Figure 1–4. The three levels at which any machine carrying out an information-processing task must be understood.

Marr, Vision (1982)

A central element in Marr's view was that a higher level was largely independent of the levels below it, and hence computational problems of the highest level could be analyzed independently of understanding the algorithm that executes the computation.
For the same reason, the algorithmic problem of the second level was thought to be solvable independently of an understanding of the physical implementation.

In contrast to the doctrine of independence of computation from implementation, current research suggests that considerations of implementation play a vital role in the kinds of algorithms that are devised and the kind of computational insights available to the scientist.

It would be convenient if we could understand the nature of cognition without understanding the nature of the brain itself.
Unfortunately, it is difficult if not impossible to theorize effectively on these matters in the absence of neurobiological constraints.
The primary reason is that computational space is consummately vast, and there are many conceivable solutions to the problem of how a cognitive operation could be accomplished.
Neurobiological data provide essential constraints on computational theories, and they consequently provide an efficient means for narrowing the search space (Churchland e Sejnowski, Science, 1988).

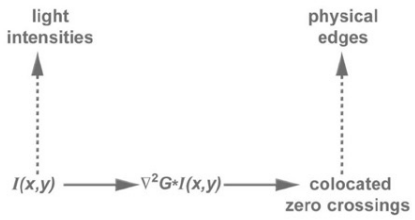


Figure 3. Formal description of edge detection. Retinal image, $I(x, y)$, is convoluted through the filtering operator $\nabla^2 G$, where G is a Gaussian and ∇^2 is a second-derivative (Laplacian) operator. Early vision processes include several filters with different Gaussian distributions, and each produces a different set of zero crossings. Intensity values are interpreted as representing light intensities in the visual field, and the colocated zero crossings are interpreted as representing edges, such as object boundaries.

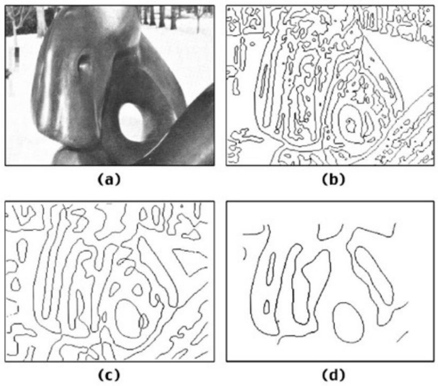
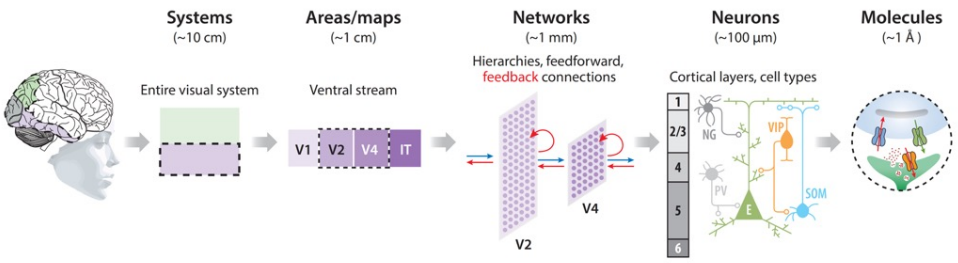


Figure 4. Different sets of zero crossings. Image (*a*) is convoluted with different-sized filters. The zero crossings obtained are shown in *b*, *c*, and *d*. Many of the fine details obtained through the smaller-sized filter (*b*) are not obtained by the larger-sized filter (*c*), but some of the zero crossings obtained in *c* do not appear in the *b*. From Marr and Hildreth (1980), 201, fig. 6. Reprinted with permission from Royal Society Publishing.

Shagrir, Philosophy of Science, 2010

Aim

Models that successfully integrate all of these levels to accurately and causally bridge from molecules to minds in visual object recognition.



Mechanistic understanding of object recognition with increasing levels of detail.

Mechanistic understanding of object recognition with increasing levels of detail.

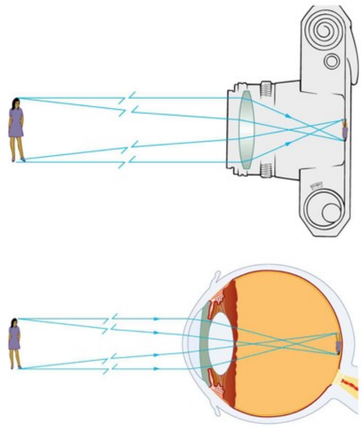
The image shows a gradual progression from the systems level (the dorsal and ventral stream of the primate visual system shown here), to areas (ventral stream areas are shown), to networks (feedforward and recurrent connectivity within the extrastriate areas V2 and V4 are shown), to neurons (a schematic of arrangement within a cortical layer is shown) and molecules (a single synapse is shown).

A thorough mechanistic understanding should gradually incorporate all levels [as suggested by Churchland & Sejnowski (1988)

Vision is often incorrectly compared to the operation of a camera.

A camera simply reproduces point-by-point the light intensities in one plane of the visual field.

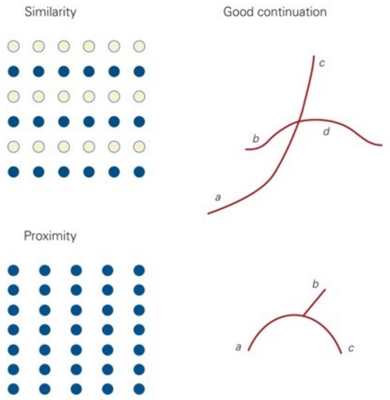
The visual system, in contrast, does something fundamentally different. It interprets the scene and parses it into distinct components, separating foreground from background.



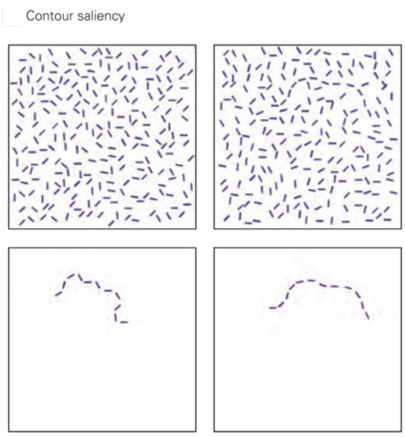
The visual system is less accurate than a camera at certain tasks, such as quantifying the absolute level of brightness or identifying spectral colors.
However, it excels at tasks such as recognizing objects (a charging animal or a speeding car) whether in bright sunlight or at dusk, in an open field or partly occluded by trees (or other cars).
And it does so rapidly (<200ms) to let the viewer respond and, if necessary, escape.

Vision is an active and bidirectional process

Vision is a generative process that involves more than just the information provided to the retina. The brain has a way of looking at the world, a set of expectations about the structure of the world that derives in part from experience and in part from built-in neuronal wiring.



To link the elements of a visual scene into unified percepts, the visual system relies on organizational rules such as similarity, proximity, and good continuation.



The principle of good continuation is also seen in contour saliency. On the right, a smooth contour of line elements pops out from the background, whereas the jagged contour on the left is lost in the background.

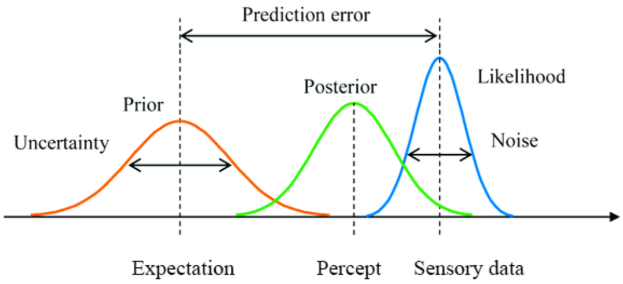
Visual priming
Higher-order representations of shape (in memory) guide lower-order processes of surface segmentation.

Bayesian theories treat the visual system as an ideal observer that uses prior knowledge about visual scenes and information in the image to infer the most probable interpretation of the image.

The posterior probability of a possible real–world stimulus S (i.e., percept) is proportional to the product of the prior probability of S (that is, the probability of S before receiving the stimulus I, e.g., expectation) and the likelihood (the probability of I given S, i.e., sensory data).

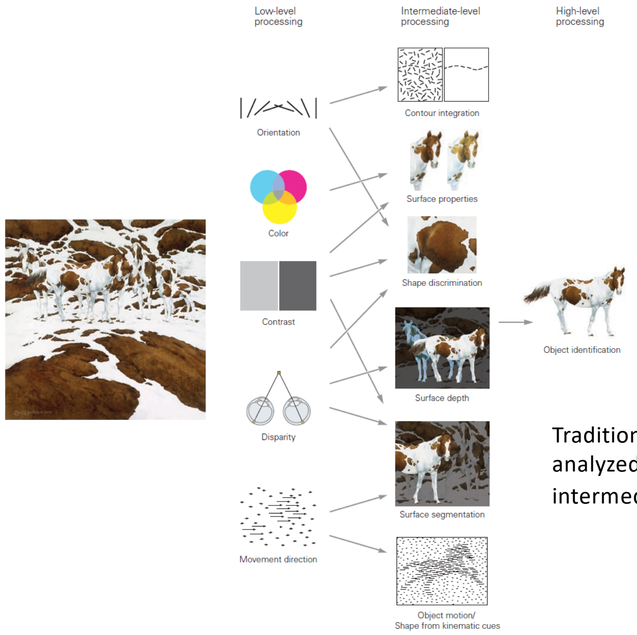
Is it reasonable to assume that the visual system knows the probability calculus and operates according to it?

$$p(S|I) = \frac{p(I|S)p(S)}{p(I)}$$



Prior probability distributions in typical applications of the Bayesian strategy represent knowledge of the regularities governing object shapes, constituent materials, and illumination, and likelihood distributions represent knowledge of how images are formed through projection on the retina. Some examples of prior knowledge are that solids are more likely to be convex than concave and that the light source is above the viewer.

The more ambiguous the image – the greater the influence of prior knowledge in yielding a nonambiguous percept.
Some perceptions may be more data-driven, others more prior knowledge driven.



Traditionally, a visual scene is analyzed at three levels: low, intermediate and high.

At the lowest level, visual attributes such as local contrast, orientation, color, depth and motion are processed.

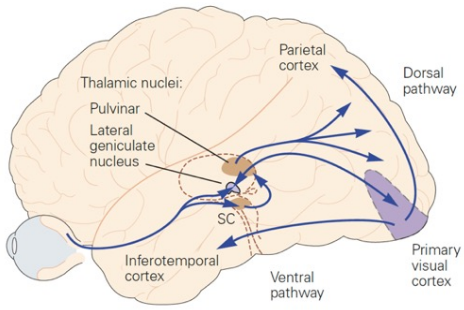
Intermediate-level processing: low-level features are used to parse the visual scene.
Local orientation is integrated into global contours (contour integration); local visual features are assembled into surfaces, objects are segregated from background (surface segmentation), surface shape is identified from depth, shading and kinematic cues.

The highest level concerns object recognition.

Once a scene has been analyzed by the brain and the objects have been recognized, the objects can be associated with memories of shapes and their meanings.

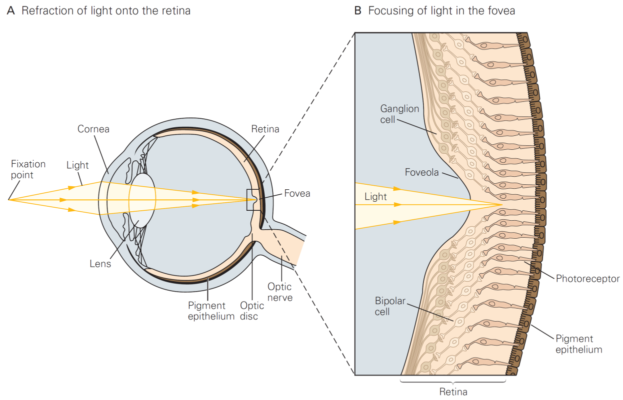
A visual scene comprises many thousands of line segments and local surface patches. Intermediate-level visual processing is concerned with determining which boundaries and surfaces belong to specific objects and which are part of the background

Visual processing is mediated by the retino-geniculo-striate pathway



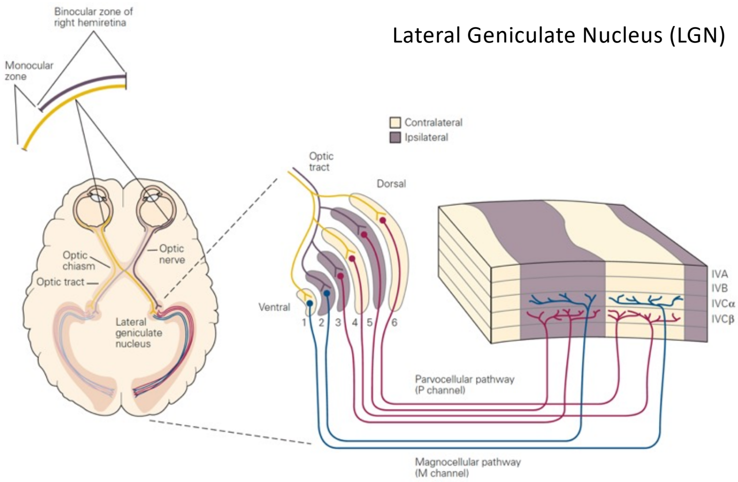
This pathway includes:

- Retina;
- Lateral geniculate nucleus (LGN) of the thalamus;
- Primary visual cortex (V1) or striate cortex;
- Extrastriate visual areas

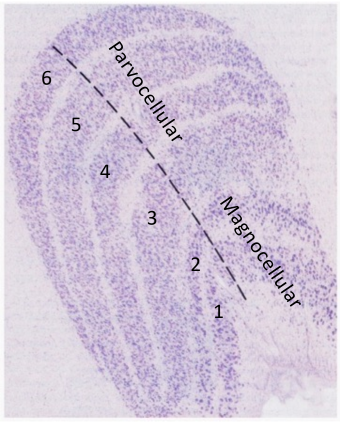


The brain's analysis of visual scenes begins in the two retinas, which transform the visual input into neural signals, a process known as phototransduction.

Lateral Geniculate Nucleus (LGN)



Beyond the optic chiasm, the axons from nasal and temporal hemiretinas carrying input from one hemifield join in the optic tract, which extends to the LGN of the thalamus.



Interspersed between the Magno and Parvo layers, are thin but dense layers called Koniocellular which receive from the K retinal ganglion cells.

Lateral Geniculate Nucleus (LGN)

In primates, LGN is a layered structure consisting of six layers, of which two Magnocellular layers (layers 1 and 2), and four Parvocellular layers (layers 3 to 6).
Each layer receives input from either the ipsilateral eye (temporal hemiretina, layers 2, 3, 5) or the contralateral eye (nasal hemiretina, layers 1, 4, 6).

Since each layer contains a map of the contralateral hemifield, the six maps are stacked on top of each other and in spatial register.

The magnocellular layers project to the IVCα layer while the Parvocellular layers project to the IVCβ layer of V1. The Koniocellular intercalated layers project to the blobs (layers 1-2) of V1.

The parallel channels established in the retina remain anatomically segregated even in LGN.

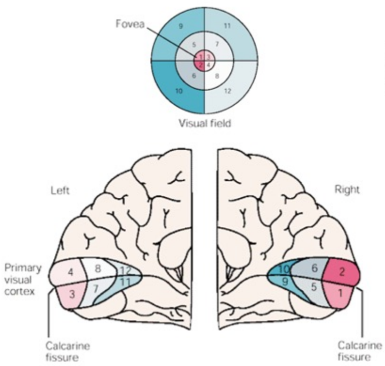
Parvocellular layers receive input from midget retinal ganglion cells, which are most numerous in the primate retina (~ 70%) and carry red-green contrast information.

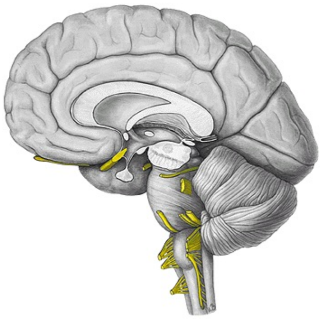
The magnocellular layers receive input from parasol retinal ganglion cells (~ 10%), which carry achromatic contrast information;

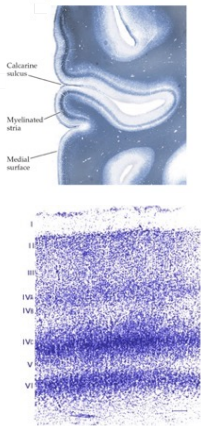
Finally, the Koniocellular layers receive input from retinal ganglion cells K (~ 10%), which carry blue-yellow opponent information;

Primary visual cortex (V1)

In humans, V1 (BA17) is located in the occipital portion of the brain along the calcarine fissure of the brain.
V1 constitutes the first level of cortical information processing.







Nissl stain of V1 reveals the different layers quite clearly.

The name striate cortex derives from the Gennari (1782) stripe, a distinctive line (visible to the naked eye) composed of terminations of myelinated axons from the LGN entering into to the IV cortical layer of V1.

The primary visual cortex is a koniocortex (sensory cortex) divided into six distinct functional layers, numbered 1 to 6.
~~Layer 4, which receives most of the visual inputs from the LGN, is in turn divided into four layers, called 4A , 4B, 4Cα, and 4Cβ.~~
~~Sublayer 4Cα receives magnocellular inputs from the LGN, while 4Cβ receives parvocellular inputs.~~

Visual area V1 has a visuotopic map of the visual field. For example, the upper bank of the calcarine fissure responds to the lower half of the visual field, the lower bank responds to the upper half of the visual field.

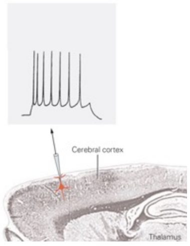
In humans, V1 contains about approximately 140 million neurons per hemisphere (Wandell, 1995), i.e. about 40 V1 neurons per LGN neuron.

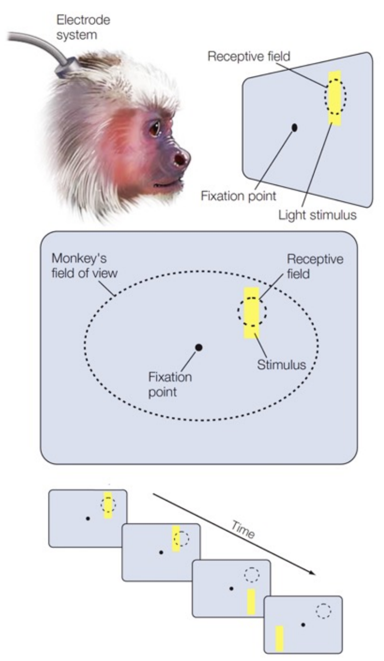
Single-cell recording

This technique allows recording signals (firing rate) from single neurons.

A fine-tipped, usually metal (platinum), electrode is inserted in the animal brain to record extracellularly change in electrical activity called action potential (AP, 1ms duration) or spike. Collected signals are appropriately amplified, filtered, viewed through an oscilloscope, and saved to a computer for offline analysis.
Since spikes are all-or-none highly stereotyped signals, most information is encoded in the brain as neuron firing rate, i.e., the number of AP in 1s.
The primary goal of single-cell recording experiments is to determine what experimental manipulations produce a consistent change in the firing rate of an isolated neuron.

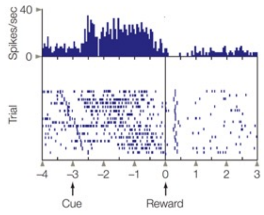
Disadvantages
– invasive
Advantages
– high spatial and temporal resolution
– differentiation between excitation and inhibition



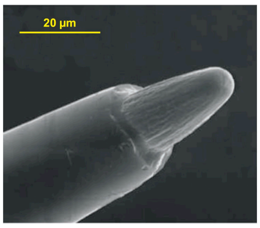


Receptive field

In early visual areas, neurons respond to stimuli in only a limited region of space.
Th is region of space is referred to as that cell's receptive field (RF).
Monkey is required to maintain fixation, and stimuli are presented at various positions in the field of view.
The neuron below fires vigorously only when the stimulus is presented in the upper right quadrant, thus defining the upper right region as the RF for this cell.



The concept of RF was introduced in 1906 by Charles Sherrington. The receptive field is a characteristic of all neurons and, in vision, indicates the region of the visual scene where the stimulus must fall to excite or inhibit the neuron being studied.



Recording of the activity of a V1 neuron. In the receptive field a moving stimulus is presented, directed downward or upward. Note that neuron firing rate is higher when the stimulus moves up.

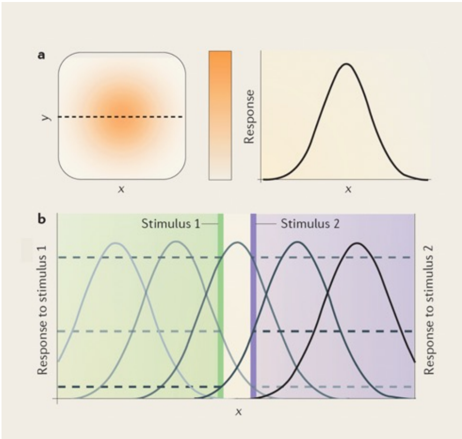
Why studying the nonhuman primate model?

First, it has human-level perceptual capabilities

Second, it allows high-spatial- and -temporal-resolution neural measurements and targeted causal perturbations to interrogate brain circuits, not feasible in humans.

Third, it has evolutionary proximity and established brain-area homologies.

Kar & DiCarlo, Ann Rev. Vis. Sci., 2024



The receptive field (RF; Charles Sherrington, 1906) of a neuron is defined as the part of stimulus space within which a stimulus elicits a response from the neuron.

In the visual system, the neuron responds to a stimulus presented in a region of space in the visual field (that is, its RF) but not to the same stimulus when it is presented outside this region.

A given stimulus typically elicits the strongest response from the centre of the RF, with the response gradually declining as the stimulus is presented further away from the centre of the RF.

Thus, the RF can be well described by a two-dimensional Gaussian distribution.

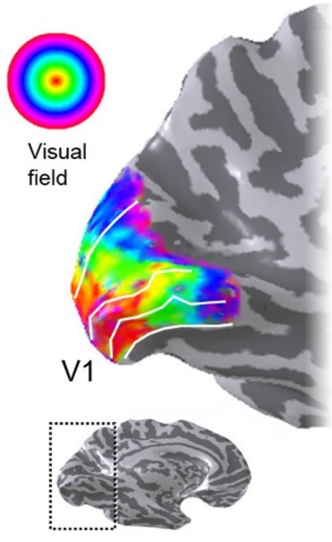
Figure b, RF profiles of a set of five neurons with different RF locations. Horizontal dashed lines indicate the response of these five example neurons to two stimuli at nearby locations (vertical green and purple lines). Both stimuli fall into the same RF (middle grey curve), but they stimulate neurons with neighbouring RFs differently so that the population can resolve the two locations even though a single neuron cannot. In addition, the size of the RF determines the neuron’s spatial frequency tuning: the smaller the RF, the higher the spatial frequency it can resolve.

Retinotopy

In early visual areas (e.g., V1 to V5), neuron RFs reveal an ordered organization, termed a retinotopic or visuotopic map.

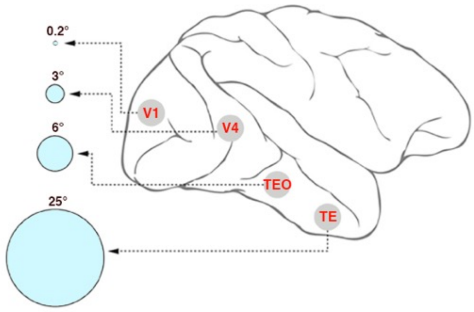
This refers to the existence of a non-random relationship between the position of neurons in the visual areas.

Neuron RFs form a 2D map of the visual field, such that neighbouring regions in the visual image (and therefore on the retinal surface) are represented by adjacent regions of the visual cortical area (i.e., orderly mapping of RF positions in retinotopic coordinates)



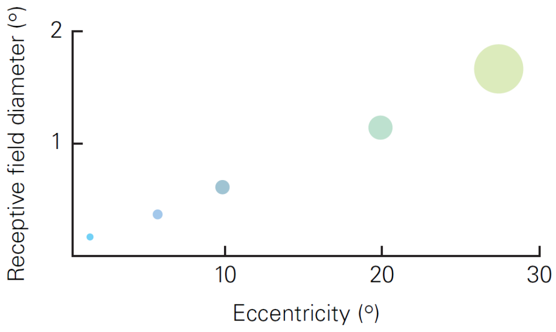
Organization of the RFs the visual system

The size of the RF changes according to the location along the visual system hierarchy



Eccentricity

The receptive fields of the retinal ganglion cells that monitor portions of the fovea subtend about 0.1 ° (equal to 6 min of arc), while those in the visual periphery reach up to 1 ° of visual angle or more.



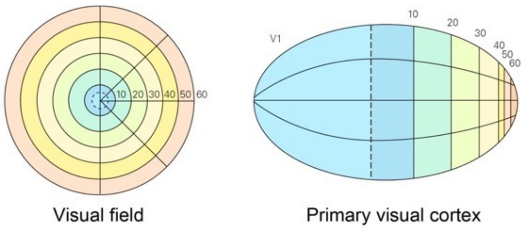
1 Arc min = 1/60 degree

The amount of cortex devoted to one degree of viewing angle changes with eccentricity.

Accordingly, more cortical space is dedicated to the central part of the visual field, where the receptive fields are smaller and densely packed and the visual system has the highest spatial resolution.

Cortical magnification

The amount of cortical area devoted to each degree of the visual field, known as the magnification factor, varies with eccentricity (i.e., the neural maps of the visual field are not isometric).
In fact, the central part of the visual field controls the largest area of the cortex.
For example, in V1 more cortex is dedicated to the central 10 ° of the visual space than to everything else.

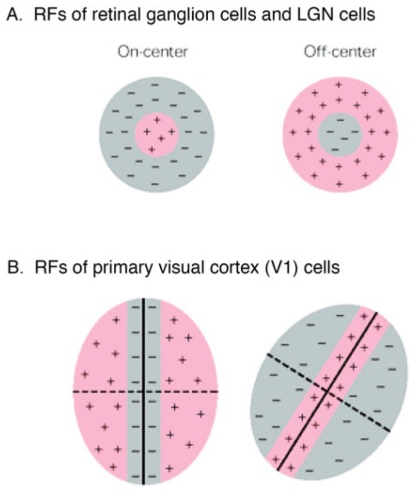


Receptive field properties

Properties change from relay to relay along a visual pathway.

By determining these properties, one can assay the function of each relay nucleus and how visual information is progressively analysed.

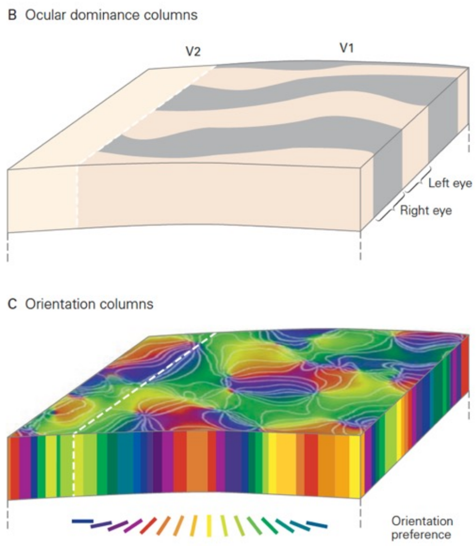
Whereas retinal ganglion cells and neurons in the LGN have concentric center-surround receptive fields, those in the primary visual cortex, although equally sensitive to contrast, analyse oriented contours (orientation selectivity).



Cortical columns

In V1, neurons with similar functional properties are found close together in columns or functional modules (about 50-75μm in diameter) that extend from the cortical surface to the white matter (about 2mm high).

V1 includes specific columns for stimulus orientation, and ocular dominance.



There are also proposals for maps of direction selectivity (Weliky et al., 1996; Ohki et al., 2005), selectivity for spatial frequency (Issa et al., 2000) and for disparity (Kara and Boyd, 2009).

Orientation columns

Neurons with the same orientation preference (i.e., vertical) are grouped together into orientation columns. Each column contains a few hundred cells and is 50-75μm wide;

Moving from one column to the adjacent one, orientation preference changes systematically by 10-15 °, both clockwise and counterclockwise, completing a 180 ° cycle (12 steps) every 750-1000μm.

The set of columns corresponding to a complete sequence of orientations (a period) is called a hypercolumn.

There are approximately 3-4 thousand hypercolumns, each monitoring a position of the visual field, in accordance with the retinotopic topology.

Hubel & Wiesel (1962, 1968) observed that when they advanced a microelectrode along a penetration perpendicular to the cortical surface, successively recorded cells shared an identical orientation of their receptive field axis.

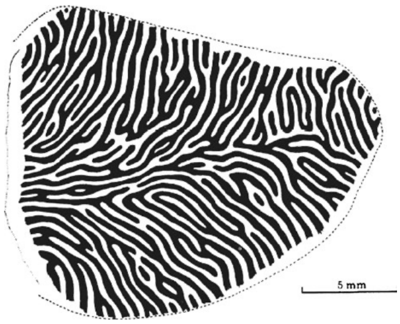
By contrast, when they advanced a microelectrode along a penetration tangential to the cortical surface, they observed that the preferred orientation of cells shifted in steps of 10-15° approximately every 65 micron.

Ocular dominance columns

The ocular dominance columns group neurons that respond more vigorously to stimuli presented to one of the two eyes. They are stripes with an average width of approximately 750 µm, running tangentially for various mm.

The ocular dominance columns reflect the segregation of inputs from different layers of the LGN, which receive inputs from retinal ganglion cells located in the ipsilateral or contralateral retina.

In tangential penetrations of V1, the dominance columns of the left and right eye have been found to alternate regularly with a periodicity of 750 to 1,000µm.



Ocular dominance columns in primary visual cortex (V1) of macaque monkey shown in tangential section. Regions receiving input from one eye are shaded black and regions receiving input from the other eye are unshaded. The dashed line signifies the border between areas V1 and V2 (taken from Hubel and Wiesel, 1977).

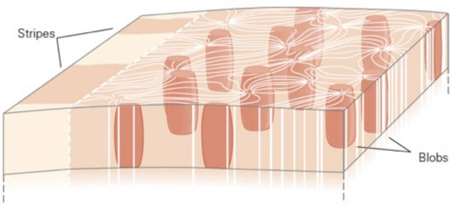
The function of ocular dominance columns remains an enigma.
One candidate function for ocular dominance columns has been stereopsis (binocular vision).
However, it has been reported that squirrel monkeys, which lack ocular dominance stripes have a stereoacuity comparable to that of human observers (Livingstone et al. 1995).

Blob e interblob

The columns of orientation and ocular dominance include groups of neurons that are poorly selective for orientation (they have circular receptors) but with strong preferences for the color of the stimulus.

These cell groups are located in the superficial layers (II and III) of V1. They are detectable by a specific marker for the cytochrome oxidase (CO) enzyme, which distributes in a regular pattern of regions defined as blobs (CO rich and color responsive) separated by interblob areas (CO poor and orientation responsive) .

D Blobs, interblobs (V1), and stripes (V2)

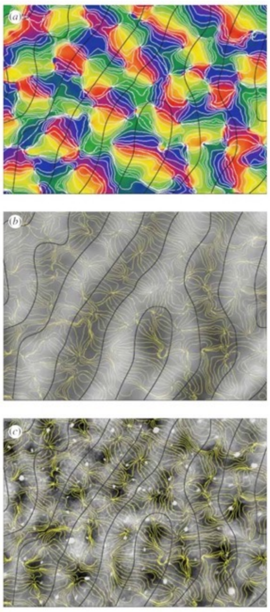


Pinwheels

More recently, optical imaging technique has enabled to visualize a surface representation of the orientation and ocular dominance columns in living animals.

The cycles of orientation columns form various structures, from parallel stripes to pinwheel-like shapes.

Sharp jumps or singularities (discontinuities) in orientation preference occur at the pinwheel centers and cause “fractures” in the orientation map.



Orientation columns

Ocular dominance columns

Blobs

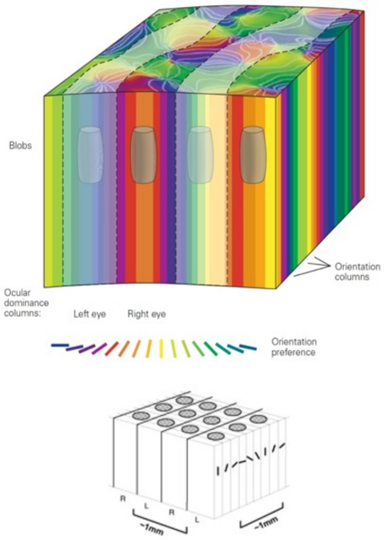
Developed for studies of cortical organization by Amiram Grinvald, this technique visualizes changes in surface reflectance associated with the metabolic requirements of active groups of neurons, known as intrinsic signal optical imaging, or changes in fluorescence of voltage-sensitive dyes.

(a) Optical imaging map of orientation preference in macaque striate cortex. The black lines represent the borders of ocular dominance columns.

(b) Iso-orientation contours from (a) superimposed on an optical imaging map of the ocular dominance columns.
Note that orientation singularities tend to be situated in the centres of ocular dominance columns.

(c) Comparison between iso-orientation contours and CO patches (blobs). In some instances, singularities and blobs coincide.
(from Horton and Adams, 2005).

Ice cube model (Hubel e Wiesel, 1977)

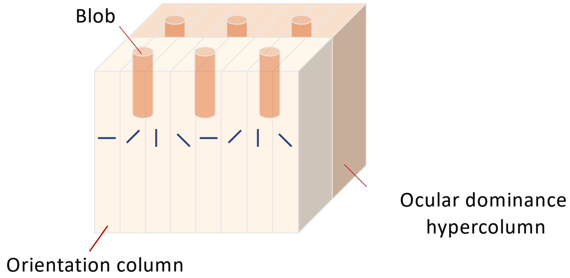


A region of cortical tissue of about 1mm contains two orientation hypercolumns (a complete cycle of selective vertical columns for orientation), one for the left eye and one for the right that alternate regularly, blob and interblob. This computational module contains all the anatomical-functional types of V1 neurons, and would be repeated thousands of times to cover the entire surface of the visual field.

However, it remains unclear what advantage, if any, is conveyed by this form of columnar segregation.
One candidate function for cortical columns is the minimization of connection lengths and processing time, which could be evolutionarily important;

The functional organization of the primary visual cortex is therefore based on two systems running orthogonally to each other:

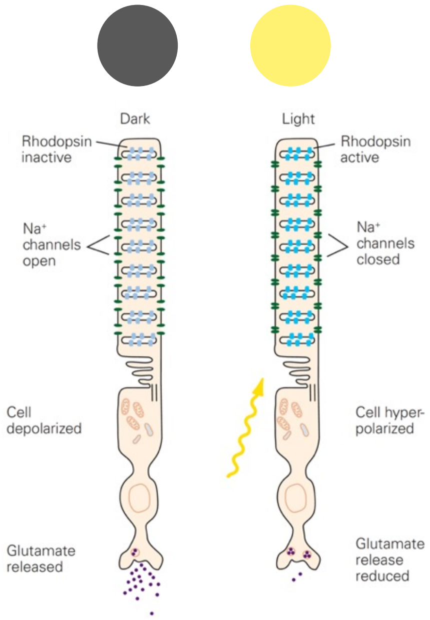
↓ orientation system
→ ocular dominance system

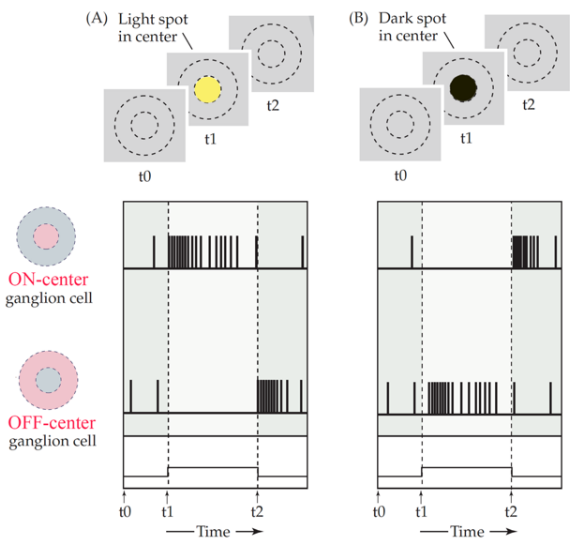


Single neurons in the visual system

Photoreceptors produce a relatively simple neural representation of the visual scene:

Neurons in the bright regions are hyperpolarized, while those in the dark regions are depolarized.





RGCs have concentric circular RFs and fall into one of two categories:

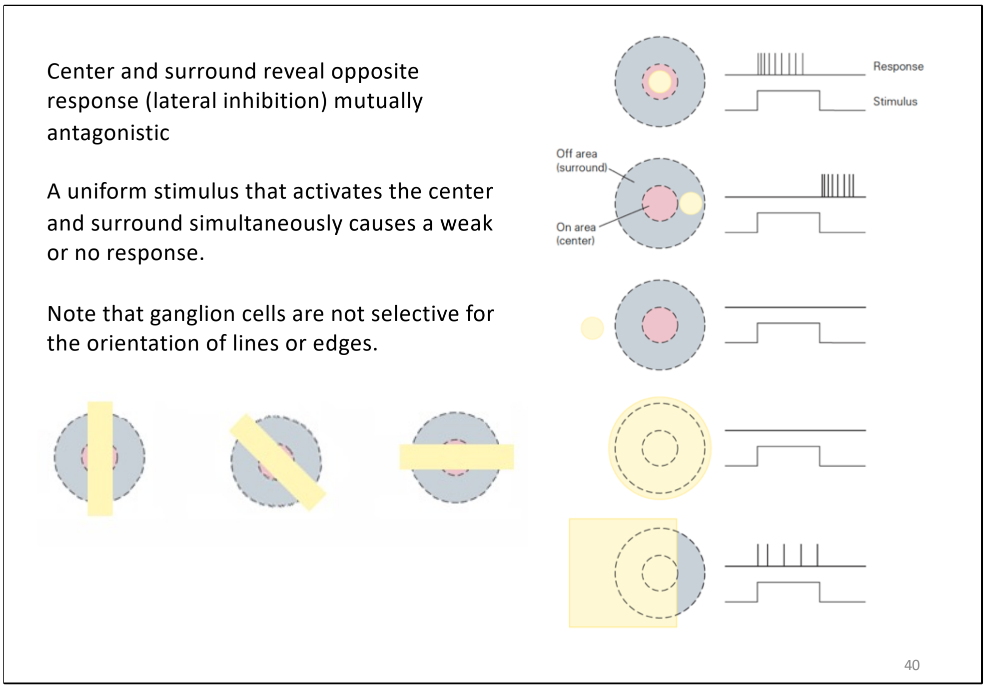
ON-center
OFF-center

The ON-center cells discharge in response to the increase in brightness in the center of the receptive field, while the OFF-center cells discharge in response to the reduction of brightness in the center of the receptive field.

ON-Center ganglion cells are excited by a light stimulus in the center of the receptive field;
OFF-Center ganglion cells are excited by a dark stimulus in the center of the receptive field;

Note that the firing rate of ON-center ganglion cells increases soon after the dark stimulus disappears;

Similarly, the discharge rate of OFF-center ganglion cells increases soon after the disappearance of the light stimulus;

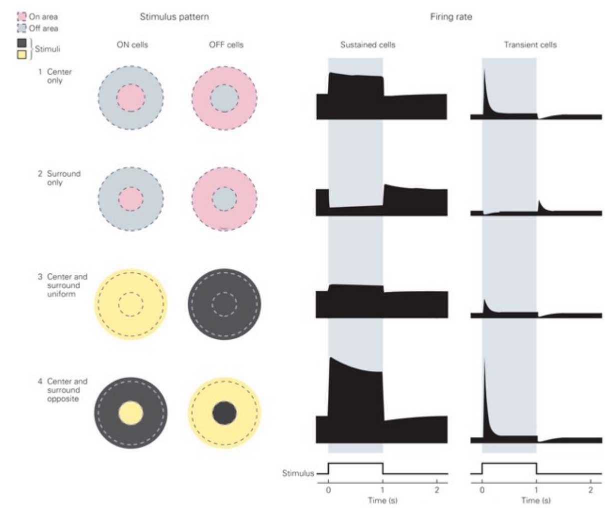


The retinal ganglion cells have an organization of the receptive field with two concentric circular areas with opposite and antagonistic response.

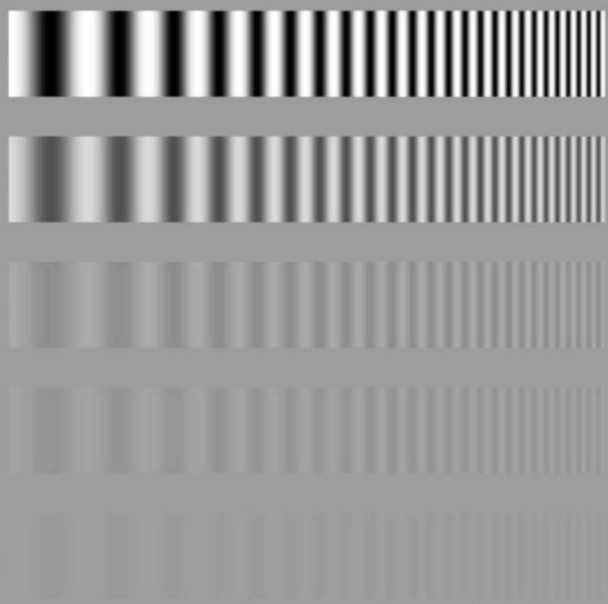
In ON-center cells, the illumination of the central part of the receptive field causes an excitatory response, i.e. an increase in the discharge of the cell, while the illumination of the surrounding part of the receptive field causes an inhibitory response (mechanism of lateral inhibition).

The OFF-center cells are instead organized in the opposite way: the illumination of the surrounding area causes an excitatory response, while the illumination of the central part of the receptive field causes an inhibitory response.

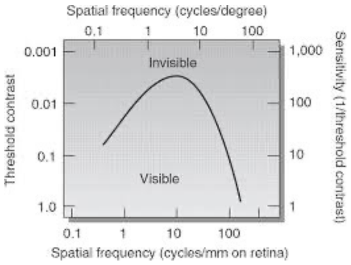
The simultaneous illumination (or darkness) of the center and surround does not evoke a variation in the discharge frequency.



Retinal ganglion cells respond weakly to uniform illumination but strongly emphasize brightness contrasts, as these carry the most reliable visual information, unlike absolute light intensity which varies with the illumination source.



Contrast perception as a function of spatial frequency



The contrast sensitivity function (CSF) describes an observer's sensitivity to sinusoidal gratings as a function of their spatial frequency.

This is measured using a contrast detection experiment in which the minimum (threshold) contrast required to detect sinusoidal gratings of various spatial frequencies is determined.

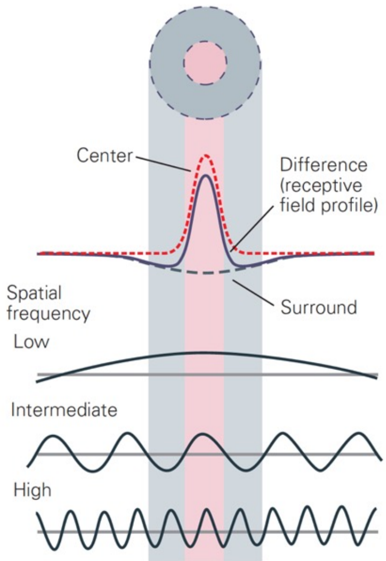
Sensitivity is defined as 1 / (threshold contrast) (so if the threshold is low, the sensitivity is high).

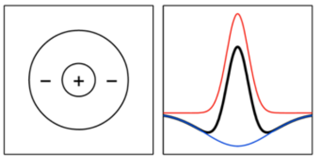
Humans are more sensitive to an intermediate range of spatial frequencies (about 4-6 cycles / degree) and less sensitive to both lower and higher space frequencies.

Gratings with a frequency of about 5 cycles per degree are the most visible. The visual system is said to have bandpass behavior because it rejects everything but a narrow band of spatial frequencies.

In the figure above, the stimulus contrast increases from top to bottom, while the spatial frequency increases from left to right.

The central bars in the figure (medium spatial frequency) are visible even at low contrast, while the wide bars and narrow bars are visible only at high contrast.





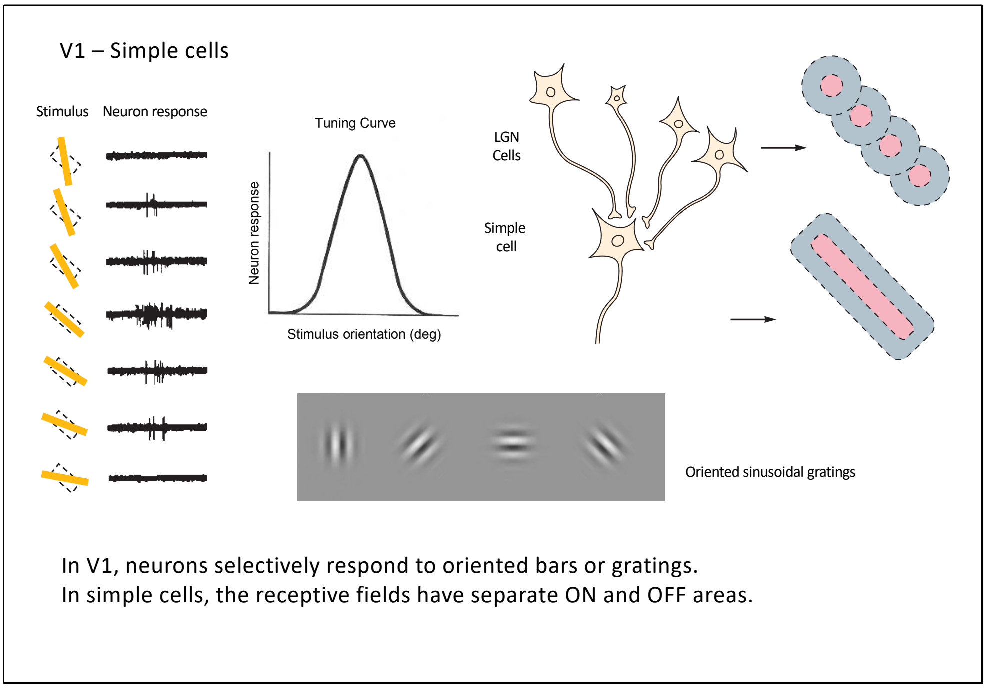
The response of the center-surround concentric RF is modeled as the difference of two Gaussian curves, a positive with a narrow base corresponding to the center of the receptive field, and a negative with a wide base, corresponding to the center and surround.

In humans, if sinusoidal gratings are used, sensitivity is greater for spatial frequencies around 5-8 cycles / visual degree, and is attenuated both for higher frequencies (up to acuity around 30-50 cycles / degree) and for frequencies less than 1 cycle / degree.

Multiplying the profile of the grating stimulus (intensity vs position) with the profile of the receptive field (sensitivity vs position) and integrating over all space calculates the stimulus strength delivered by a particular grating.

In day light, contrast sensitivity declines sharply at high spatial frequencies, with an absolute threshold at approximately 50 cycles per degree.

Interestingly, sensitivity also declines at low spatial frequencies. The attenuation at low frequencies reflects the inhibitory and antagonistic action of the periphery (surround) of the receptive fields of the retina, geniculate and cortex.
Patterns with a frequency of approximately 5 cycles per degree are most visible.
The visual system is said to have band-pass behavior because it rejects all but a band of spatial frequencies.

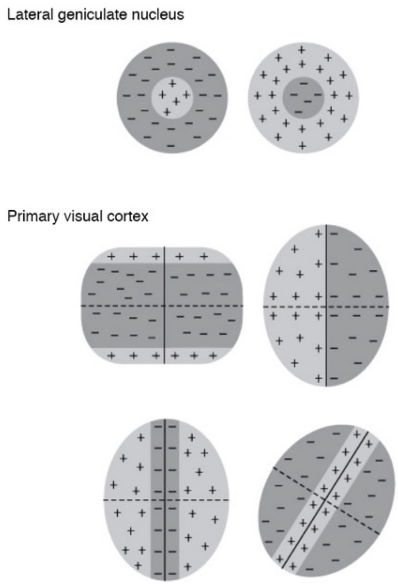


Neurons in area V1 are classically divided into two types: simple and complex (Hubel and Wiesel, 1959).
Neurons have elongated RFs and respond to a narrow range of orientations.
Different neurons respond optimally to distinct orientations (orientation tuning curve).
Example of a neuron in area V1 that selectively responds to lines that adapt to the orientation of its receptive field.
This selectivity is the first step in the brain's analysis of the shape of an object.

The orientation of the receptive field is thought to result from the alignment of the center-surround circular receptive fields of different LGN cells.

In the monkey, the neurons of the LGN have non-oriented circular receptive fields. However, projections of adjacent LGN cells onto a simple cell create a receptive field with a specific orientation.

Simple cells respond well to sinusoidal gratings (Gabor patches) of specific spatial frequencies and phases.



The receptive fields of the simple cells of the primary visual cortex are different and less homogeneous than those of the ganglion cells of the retina and the LGN

Complex cells

Have rectangular receptive fields, larger than those of simple cells;

Respond to linear stimuli with specific orientation;

The position of the stimulus within the receptive field is not critical as the demarcation between on and off zones is not so clear;

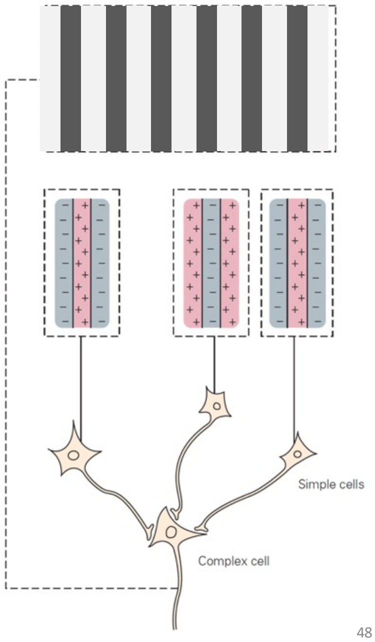
Movement of the stimulus in the receptive field is particularly effective in activating the cells;

Complex cells selectively respond to stimuli that move in particular directions;

V1 – Complex cells

In complex cells, the ON and OFF regions are superimposed, i.e. each position in the receptive field responds to both white and black bars, and the cells respond when a line or edge crosses the receptive field along an axis perpendicular to the orientation of the receptive field.

This constancy in the response to variations of stimulus location in the RF is commonly called position invariance.



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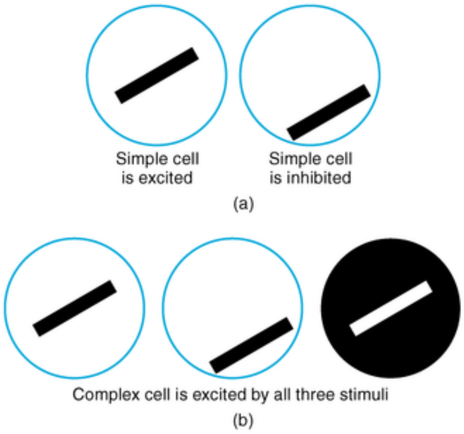
Complex cells are less selective for the position of the stimulus in the receptive field

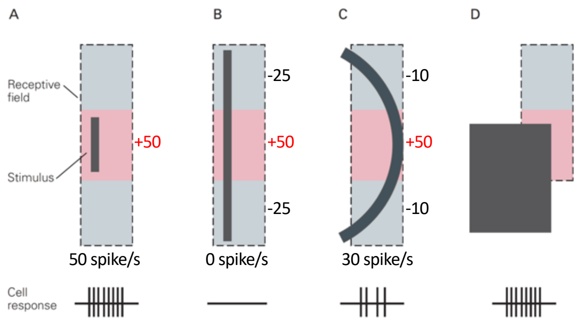
The receptive field has no defined ON and OFF regions and responds similarly to light (on a dark background) or dark (on a light background) stimuli in all positions of the receptive field.

They are activated as a linear oriented stimulus crosses their receptive fields in one direction.

► **Response Characteristics of Neurons to Orientation in the Primary Visual Cortex**



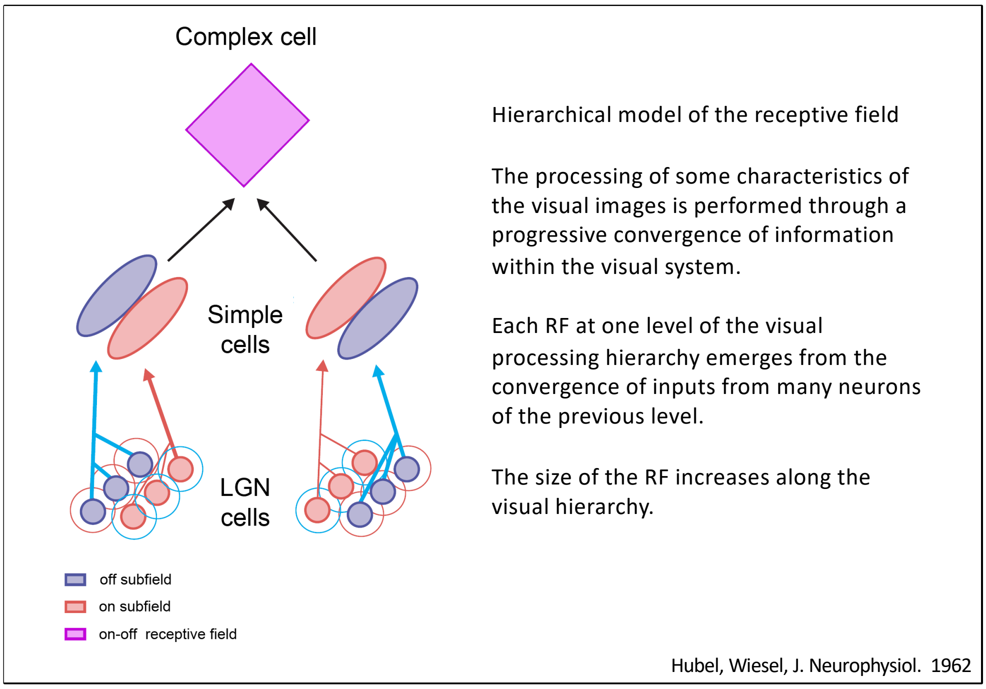
End-stopped cells



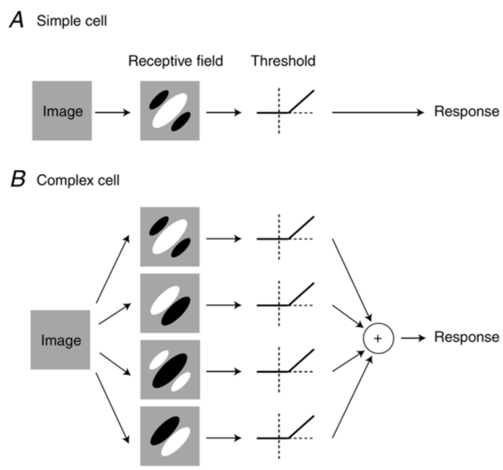
The receptive fields of some cells have a central excitatory region flanked by inhibitory regions that have the same preferential orientation.
A short linear segment (A) or a long curved line (C) will be effective in activating the neuron, but not a long straight line (B).
A neuron with a receptive field that has only one inhibitory region in addition to the excitatory region can signal the presence of angles (D).

Respond better to linear stimuli of a certain length, or that have an end that does not extend beyond a specific portion of the cell's receptive field.
End-stopped may serve to detect angles ("angle-detectors") or curved lines of visual images.

ON and OFF regions of the RF have the same preferred orientation (vertical, in the neuron illustrated in the figure).
Therefore, the inhibitory effect is greater if the same oriented contour is presented both in the ON and OFF regions.
A short linear segment (A), or a long curved line (C) will be effective in activating the neuron, because excitation will be greater than inhibition.
On the contrary, a long straight line (B) will not be effective, because excitation will be canceled by the inhibitory effect.



According to the hierarchical model (Hubel and Wiesel, 1962), simple cell receptive fields are constructed from the convergence of geniculate inputs with receptive fields aligned in the visual space.
In turn, complex receptive fields arise from the convergence of simple cells with similar orientation preferences.



The models of simple and complex cells proposed by Movshon, Thompson and Tolhurst (Movshon et al. 1978)

A, simple cells. The first stage is linear filtering, i. e. a weighted sum of the image intensities, with weights given by the receptive field. The second stage is rectification: only the part of the responses that is larger than a threshold is seen in the firing rate response.

B, complex cells. The first stage is linear filtering by a number of receptive fields such as those of simple cells (here we show four of them with spatial phases offset by 90 deg). The subsequent stages involve rectification, and then summation.

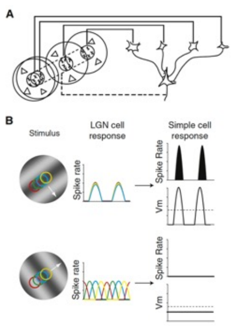
V1 complex cells in the primary visual cortex exhibit a degree of spatial and phase invariance to edges within their RF. This means they can respond to specific orientations of edges regardless of exact position within a certain area.

Complex cells are thought to receive input from multiple simple cells (pooling), each of which is tuned to: the same orientation, lightly different positions within the visual field. Complex cells lose phase sensitivity by combining input from simple cells with different preferred phases.

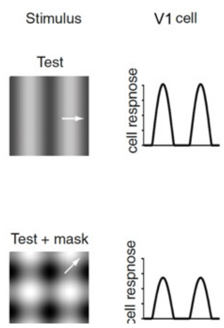
This makes them particularly good at detecting edges and contours regardless of their precise position—a key step in building up more invariant object recognition in the visual system.

A complex cell pools the (rectified) outputs of several simple cells. This creates spatial and phase invariance to edge location within a certain range.

The feedforward model of orientation selectivity in V1



The feedforward model as originally proposed by Hubel and Wiesel (1962). Four cells from the LGN, whose receptive fields are shown to the left, synapse onto a V1 simple cell. The simple cell derives its preferred orientation from the axis of alignment of these cell receptive fields.



Cross-orientation suppression cobtradicts a purely feedforfward model of orientation swlectivity in V1

A neuron in V1, especially a complex cell, might respond strongly to a grating at its preferred orientation. However, if you superimpose a second grating at a non-preferred (orthogonal) orientation, the response of the neuron often decreases, even though this second grating by itself doesn't excite the neuron.

The classic feedforward model predicts that:

A V1 complex cell pools inputs from simple cells tuned to the same orientation The cell's output depends mainly on the presence of its preferred orientation, largely ignoring orthogonal orientations (because they don't activate the relevant simple cells).

But cross-orientation suppression shows that:

A stimulus with an orthogonal orientation — even if not activating the preferred circuit — can still inhibit the response

This suggests extra mechanisms at work, beyond just the feedforward pathway

Normalization via Inhibitory Circuits

The cortex performs divisive normalization: the response of a neuron is divided by the total activity in a population. So even if orthogonal stimuli don't directly excite the neuron, they boost overall local activity, leading to suppression. This is a nonlinear computation and likely involves lateral or feedback inhibition

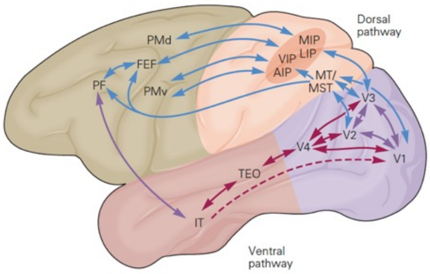
Cross-orientation suppression shows that V1 neurons are not just simple feature detectors passively summing feedforward inputs. Instead, they:

•Participate in **dynamic cortical circuits**
•Are influenced by **contextual and global properties**
•Are regulated by **nonlinear, recurrent, and feedback computations**

Beyond V1 are the extrastriate visual areas (more than 30 areas in macaques), a set of higher-order visual areas organized as neural maps of the visual field.

Visual areas are organized in two hierarchical pathways, a ventral pathway involved in object recognition and a dorsal pathway dedicated to the use of visual information for guiding movements.

The ventral or object recognition pathway extends from V1 to the temporal lobe
The dorsal or movement-guidance pathway connects V1 with the parietal lobe and then with the frontal lobes.



Ungerleider & Mishkin, Two cortical visual systems. 1982

The dorsal and ventral pathways are highly interconnected so that information is shared.

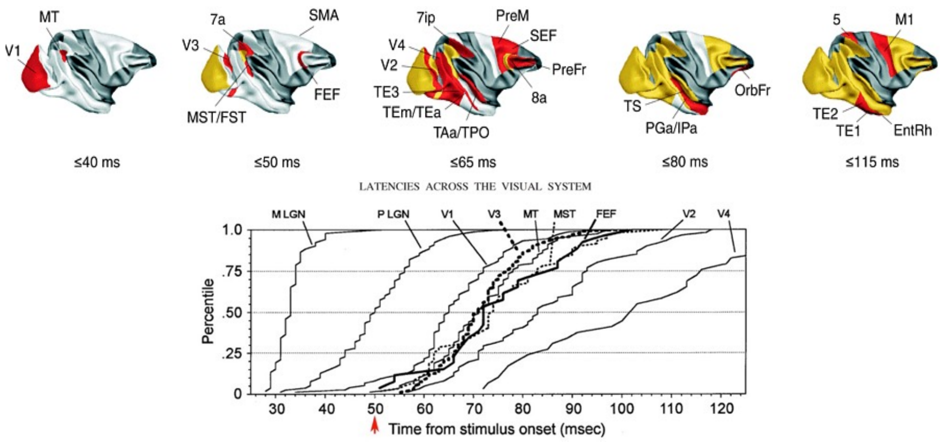
For example, stimulus movement information in the dorsal pathway (area V5) can contribute to object recognition through kinematic cues. Information about movements in space derived from areas in the dorsal pathway is therefore important for the perception of object shape and is fed into the ventral pathway.

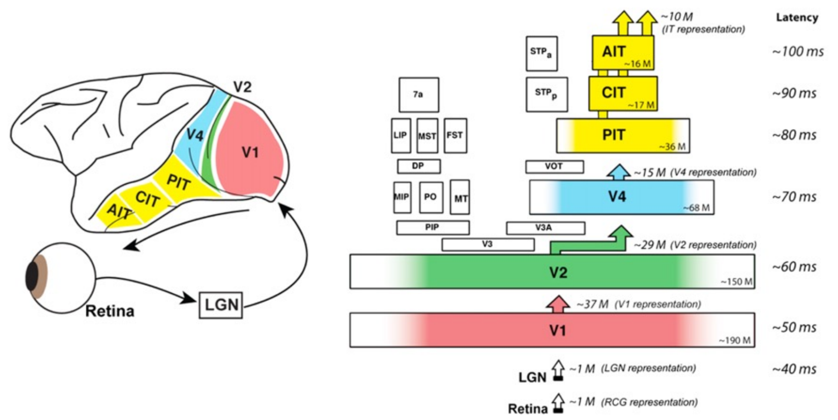
Note: all connections between areas in the ventral and dorsal pathways are reciprocal: each area sends information to the areas from which it receives input.
Reciprocity is an important feature of connectivity between cortical areas

Ventral vs. Dorsal Visual Pathways

Aspect	Ventral Pathway ("What")	Dorsal Pathway ("Where/How")
Function	Object recognition, identification	Spatial awareness, movement coordination
Pathway	Occipital → Inferior Temporal Cortex	Occipital → Posterior Parietal Cortex
Main Input Type	Parvocellular (local, color)	Magnocellular (global, motion)
Processes	Shape, color, textures	Motion, depth, position
Consciousness	Linked to conscious perception	Often unconscious, guides real-time action
Damage Effects	Visual agnosia (can't identify objects/faces)	Optic ataxia (impaired visually guided movement)
Example	Recognizing a face	Grasping a moving object

Visual information in both the ventral ("what") and dorsal ("where/how") pathways is processed hierarchically in the brain. As signals progress through each stage, neuronal responses become slower, receptive fields get larger, and stimuli become more complex—reflecting more abstract levels of processing.





Visual area V4 is an intermediate cortical area in the ventral visual pathway.

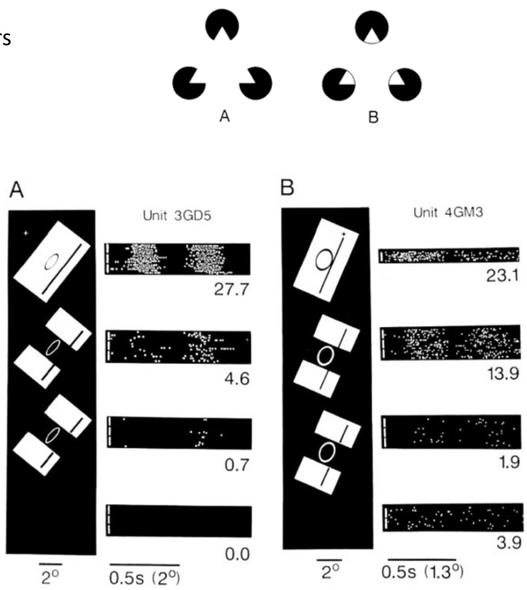
It considered to be crucial for visual object recognition and visual attention.
A primary role of V4 is to facilitate figure-ground segmentation of the visual scene, thus enabling both bottom-up and top-down visual processes.

In the macaque monkey, V4 is located on the prelunate gyrus and in the depths of the lunate and superior temporal sulci and extends to the surface of the temporal-occipital gyrus.

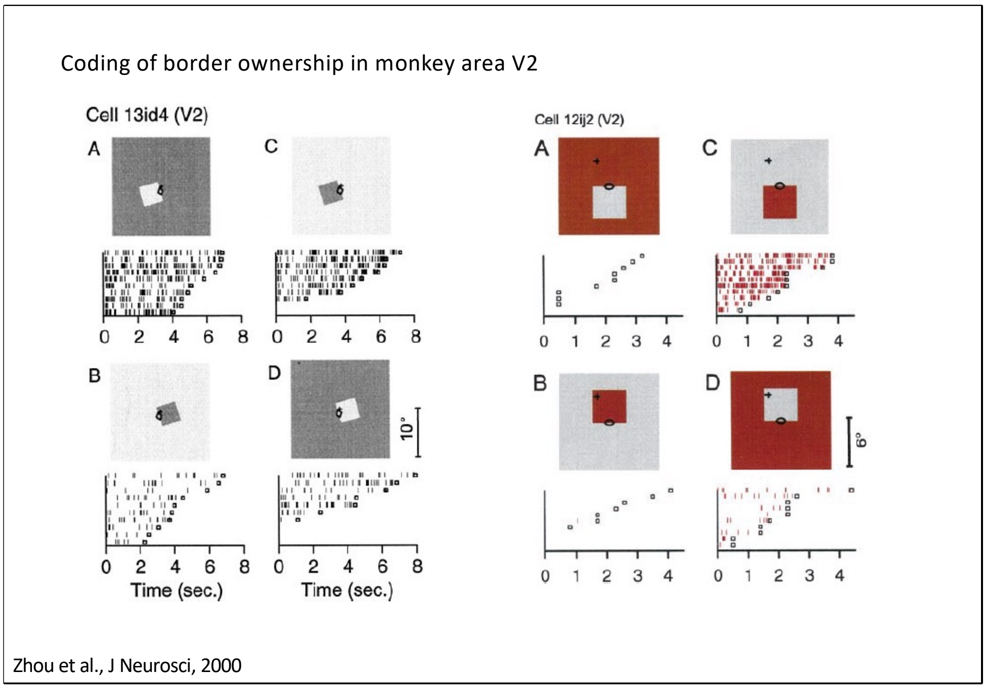
Neural responses to illusory contours

Figures in which humans perceive illusory contours evoke responses in cells of area 18 (V2) in the monkey visual cortex. Modifications that weakens the perception of contours also reduces the neuronal responses.

In contrast, cells in V1 (area 17) were apparently unable to see these contours.



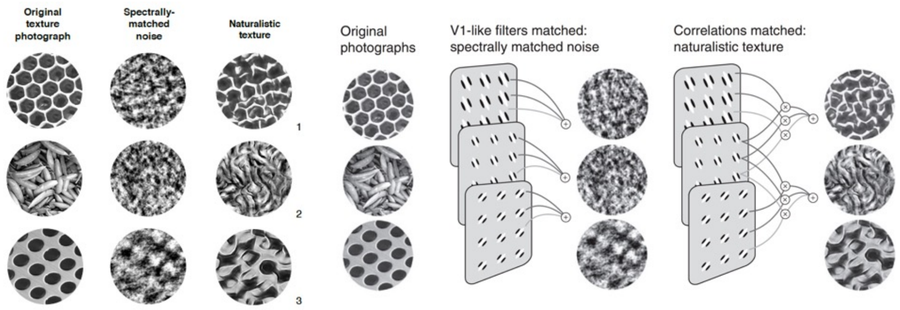
Von der Heydt et al., Science, 1984



The influence of visual stimulation far from the receptive field center indicates mechanisms of global context integration.
The short latencies and incomplete cue invariance suggest that the border-ownership effect is generated within the visual cortex rather than projected down from higher levels.

There is no generally accepted account of the function of V2, partly because no simple response properties robustly distinguish V2 neurons from those in V1.

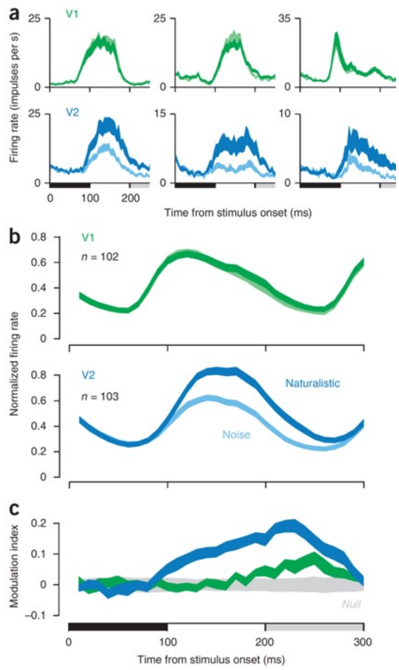
Stimuli replicating the higher-order statistical dependencies found in natural texture images were used to stimulate macaque V1 and V2 neurons.



Freedman et al., Nat. Neurosci., 2013

Spectrally matched noise images.
The original texture is analyzed with linear filters and energy filters (akin to V1 simple and complex cells, respectively) tuned to different orientations, spatial frequencies and spatial positions.
Noise images contain the same spatially averaged orientation and frequency structure as the original but lack many of the more complex features.

Naturalistic texture images. Correlations are computed by taking products of linear and energy filter responses across different orientations, spatial frequencies and positions. Images are synthesized to match both the spatially averaged filter responses and the spatially averaged correlations between filter responses. The resulting texture images contain many more of the naturalistic features of the original.



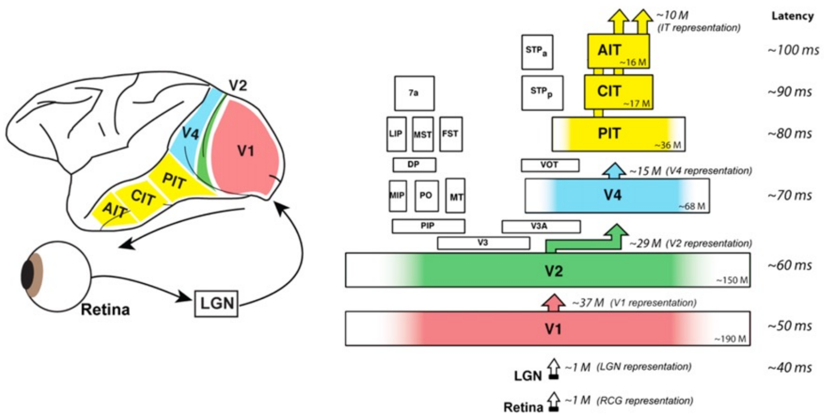
Neuronal responses to naturalistic textures differentiate V2 from V1 in macaques.

a) Time course of firing rate for three single units in V1 (green) and V2 (blue) to images of naturalistic texture (dark) and spectrally matched noise (light). Black bar indicates the presentation of the stimulus; gray bar indicates the presentation of the subsequent stimulus.

b) Time course of firing rate averaged across neurons in V1 and V2. Each neuron's firing rate was normalized by its maximum before averaging.

c) Modulation index, computed as the difference between the response to naturalistic and the response to noise, divided by their sum. Modulation was computed separately for each neuron and texture family, then averaged across all neurons and families.

Freedman et al., Nat. Neurosci., 2013

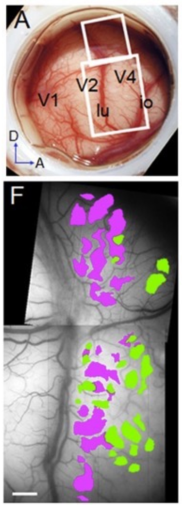


Visual area V4 is an intermediate cortical area in the ventral visual pathway.

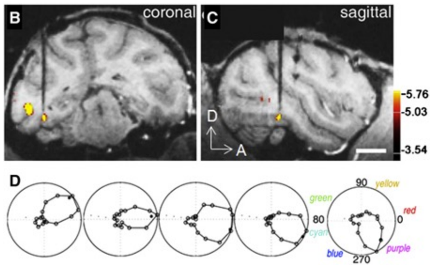
It considered to be crucial for visual object recognition and visual attention.
A primary role of V4 is to facilitate figure-ground segmentation of the visual scene.
In other words, V4 is where a representation for “things” begins to emerge.

In the macaque monkey, V4 is located on the prelunate gyrus and in the depths of the lunate and superior temporal sulci and extends to the surface of the temporal-occipital gyrus.

Joint encoding of boundary and surface properties in order to facilitate object segmentation



Recent advances in fMRI and optical imaging methods have revealed that V4 is not a homogenous area and may comprise segregated functional domains, dubbed globs (in analogy with the color blobs of V1), specialized for color processing, and interglobs, for orientation and shape processing



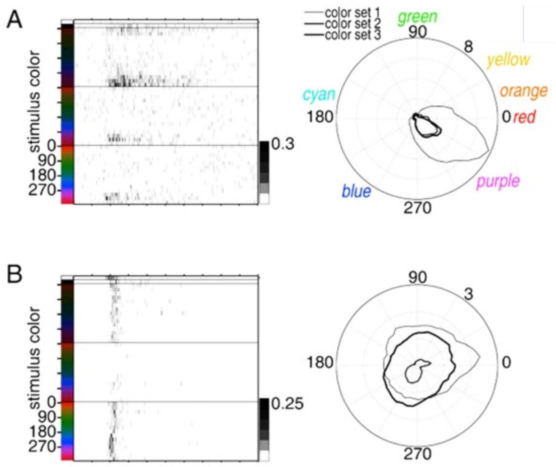
(F) illustrates segregation of color/ Lum and orientation preference bands in area V4

Several color globs identified with fMRI seen in coronal (B) and sagittal (C) views.

Color voxels are better activated by equiluminant color than black and white gratings.

Electrode is seen targeting a color glob. (D) Shift in color preference of neurons along length of an electrode penetration through a glob.

Figure shows color-response measurements for a typical cell in a glob (A), and a cell in the interglob (B) region. Left panels show time histograms to an optimally shaped bar of various colors; right panels show the color tuning in polar coordinates.



Most glob cells were excited by a specific hue, as shown by hue tuning in the polar plots.

This hue selectivity was luminance invariant.
That is, hue tuning does not change with change in luminance between stimulus and background.

Conway et al., Neuron, 2007

These measurements were conducted by presenting an optimally configured bar at the center of the response region for each cell, and in each trial changing the color of the bar.

The top two rows in each histogram show the responses to white and black.

The rest of the histogram plot is divided into three sections.
The top section shows responses to a set of colors that were equiluminant with each other but lower luminance than the background (top), equiluminant with background (middle), and of higher luminance than the background (bottom).

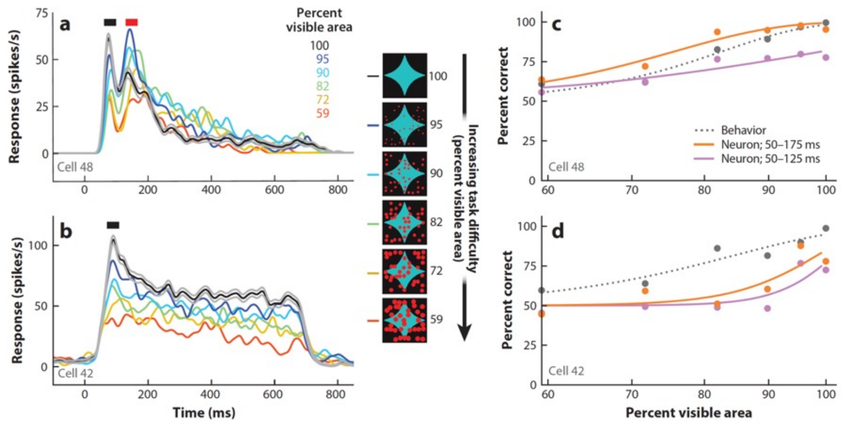
The glob cell shown in figure was excited by bluish-red, shown by the maximal response density at the bottom of each section of the histogram.

The hue responses can be compressed into polar plots. Despite differences in overall response magnitude to the three color sets, the peak hue response within each color set was the same: each of the threecurves in right panels point to 330.

This shows that the hue selectivity was luminance invariant.
Luminance invariance does not imply that luminance does not modulate the responses, just that a change in luminance does not drastically shift or cancel out the hue tuning.

Luminance invariance can be quantified by determining the degree to which the patterns of responses to the different color sets are correlated.
The correlation coefficient of the response to color set 1 and color set 2 for was 0.94; between color set 1 and 3, 0.87; and between color set 2 and 3, 0.96; average, 0.92.

The interglob cell (B), on the other hand, did not show strong hue tuning, to any color set;



Responses of two example V4 neurons that exhibit different response profiles to partially occluded shape stimuli.

Pasupathy et al., Ann Rev Vis Sci, 2020

Responses of two example V4 neurons that exhibit different response profiles to partially occluded shape stimuli.
Cell 48 in panel a exhibit two transient peaks (black and red bars), while cell 42 in panel b exhibit only one transient peak (black bar).

During the first transient in both neurons, responses decline with increasing levels of occlusion (line colors; decreasing percent visible area).

During the second transient in panel a, which may be based on feedback from frontal cortex, responses are stronger for intermediate levels of occlusion (for further details, see Fyall et al. 2017, Kosai et al. 2014).

(c, d) Psychometric and neurometric curves based on the data in panels a and b, respectively. In both, psychometric performance (dark gray dotted line) declines with increasing levels of occlusion. The neurometric curve based on a larger time window (orange line) shows improved performance at intermediate levels of occlusion for the neuron in panel a but not the one in panel b due to the enhanced shape selectivity for occluded stimuli during the second transient peak.

Object identification and categorization

The visual experience of the world is fundamentally centered on objects.

By visual object we mean a set of visual characteristics (e.g., visual features) grouped or joined perceptually in discrete units on the basis of the organizational principles of the Gestalt, such as proximity, similarity, closure, good continuation, good form, connection, etc.

By visual recognition we mean the ability to assign a verbal label (e.g., a name) to objects in the visual scene.

There are at least two possible object recognition tasks, distinguished by level of specificity: identification and categorization.

An object can be recognized at an individual level (e.g., a Siamese cat), or at a more general categorical level, as an object belonging to a given class (a cat, a mammal, an animal, and so on).

It is quite simple for computer vision techniques to identify (rather than categorize) objects.

On the contrary, for the human vision the task of identification (compared to categorization) is more difficult.

Categorization
A category exists whenever two or more distinct objects or events are treated equivalently.

For example, when distinct objects or events are labeled with the same name, or when the same action is performed on different objects.
Although the stimuli are distinct, organisms do not treat them uniquely; but they respond on the basis of past experience and categorization.

In this sense, categorization can be considered one of the most basic functions of living beings (Mervis and Rosch, 1981)

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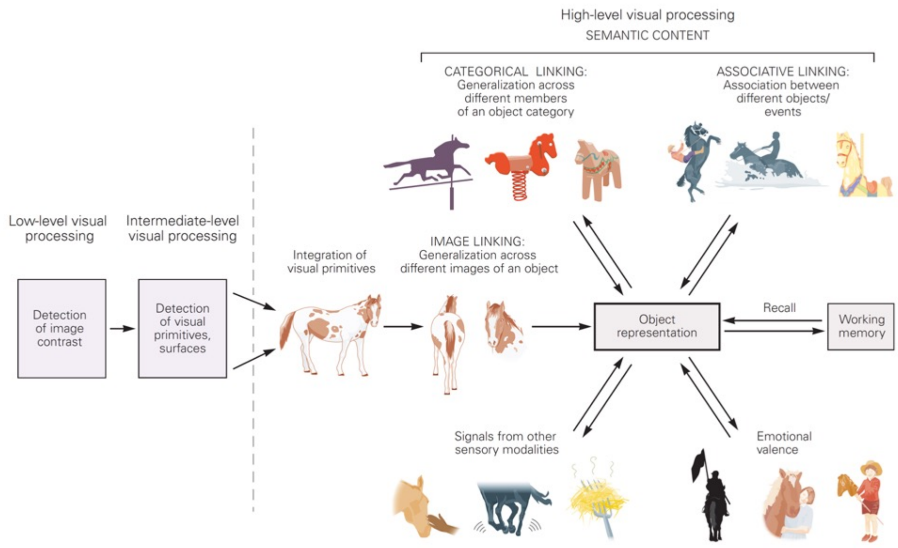
High-level visual processing is concerned with object recognition



We effortlessly and rapidly (100-200ms) detect and classify objects from among tens of thousands of possibilities despite the tremendous variation in appearance that each object produces on our eyes.

Our daily activities (e.g., finding food, social interaction, selecting tools, reading, etc.), and thus our survival, depend on our accurate and rapid extraction of object identity from the patterns of photons on our retinae.

The fact that half of the nonhuman primate neocortex is devoted to visual processing speaks to the computational complexity of object recognition.

Object recognition involves integration of visual features extracted at earlier stages in the visual pathways.

This integration requires generalization across different retinal images of an object, as well as generalization across different members of an object category.

The representation also incorporates information from other sensory modalities, attaches emotional valence, and associates the object with the memory of other objects or events.

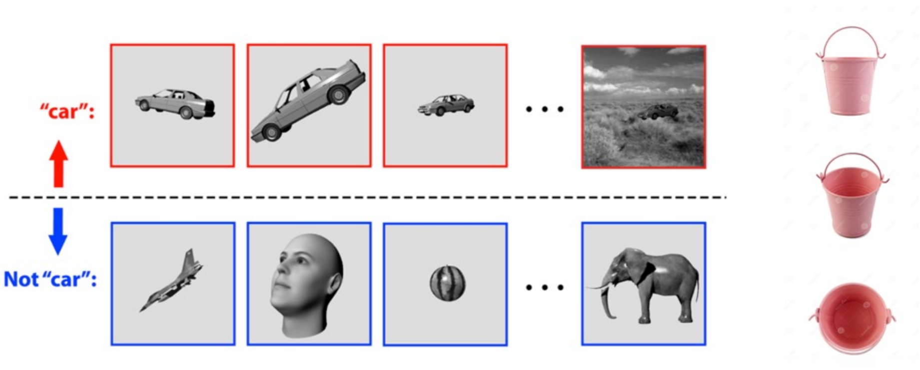
Selectivity and object constancy (or invariance)

A computational difficulty of object recognition is that it requires both:

selectivity (different responses to distinct objects, such as one face with respect to another face);

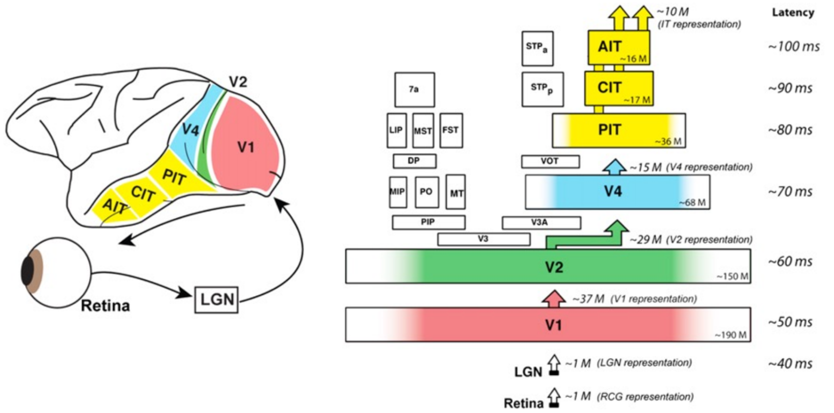
and invariance with respect to image transformations (similar responses to, for example, rotations or translations of the same face);

In fact, we are able to recognize the same object even when the image it projects on the retina varies considerably.



Core object recognition is the ability to rapidly (<200 ms viewing duration) discriminate a given visual object (e.g., a car, top row) from all other possible visual objects (e.g., bottom row).

Primates perform this task remarkably well, even in the face of (identity-preserving) transformations (e.g., changes in object position, size, viewpoint, and visual context).

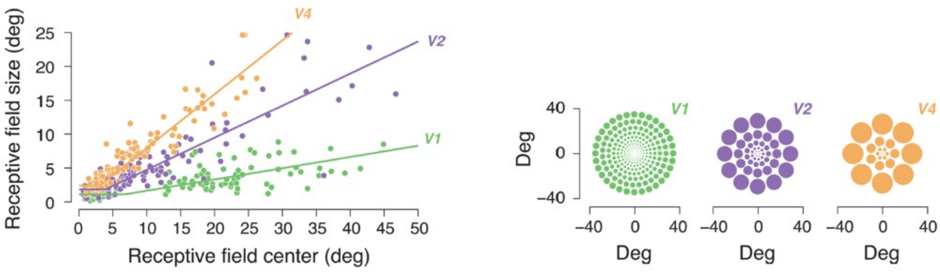


Studies in animals (e.g., primates) primarily implicates the inferior temporal cortex in object perception and recognition.

The inferior temporal cortex is a large region of the brain that includes at least two major functional subdivisions - the posterior area, or temporo-occipital cortex (TEO area), and the anterior area, or inferotemporal cortex (IT area).

Area V1,V2 and V4 are located in the occipital lobe;

Area TEO (TEmporal-Occipital) and IT (InferoTemporal) are located in the temporal lobe;



In contrast to V1, V2, V4 and partly PIT, CIT and AIT show no clear retinotopy.

Large receptive fields (from 2 to 30 degree). Most of the receptive fields include the fovea. Foveal stimuli evoke stronger responses.

Freeman and Simoncelli, Nat. Neurosci., 2013

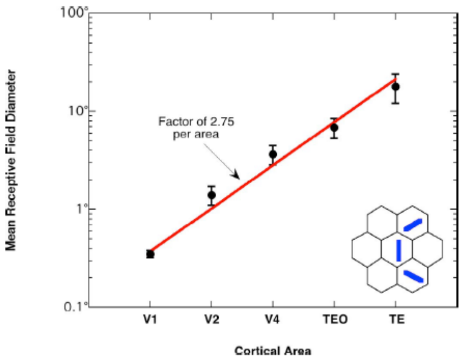
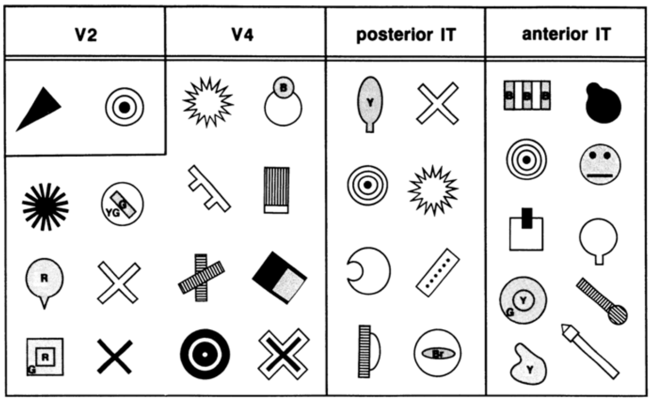


Figure 1. Mean data from four studies on receptive field diameter in successive cortical areas in the ventral pathway. Error bars represent the range of means across the four studies. The red line shows that the data can be fit by constant diameter increase of 2.75× from area to area. Inset in lower right shows an example of a curved contour represented by a combination of adjacent, oriented (orientations in blue) V1 receptive fields (individual hexagons) in a 3.0× larger diameter V2 receptive field.

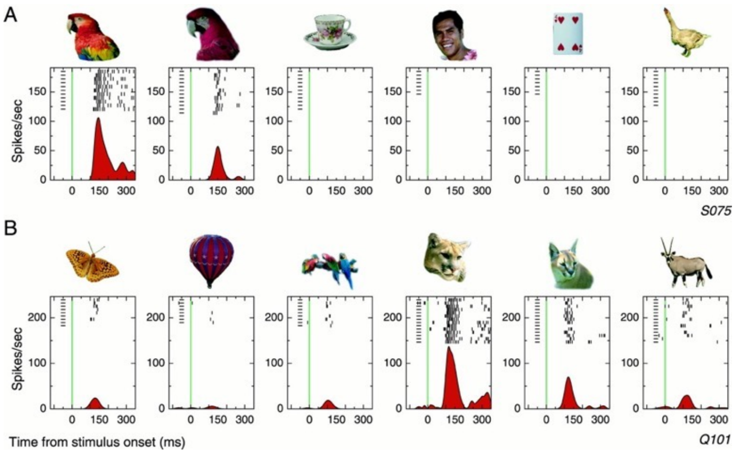
The diameter increase of a factor of about 3.0 is consistent with receptive fields one area higher in the hierarchy being constructed from combinations of nearest neighbours in its input area.

Increased complexity of effective stimuli along the ventral visual path

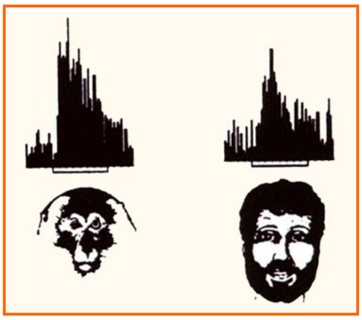


Kobatake and Tanaka, J. Neurophysiol., 1994

Neurons in the IT respond to relatively complex stimuli, often to biologically relevant objects such as human and other animal faces, hands and other parts of the body.

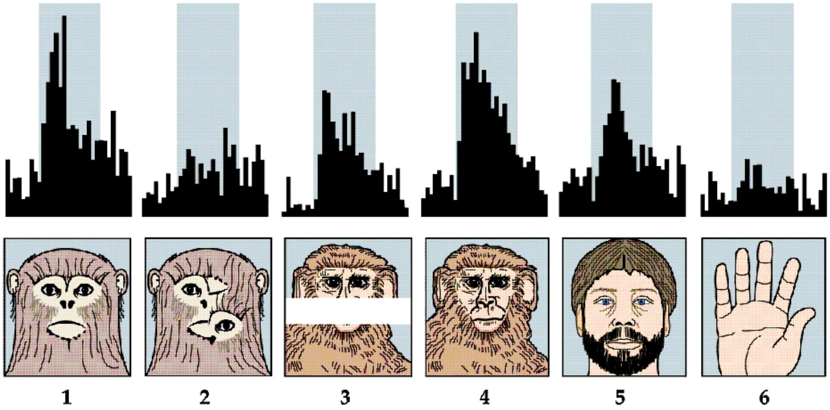


In the early 1980s, several researchers (Bruce et al., Perrett et al.) Identified in the monkey a group of IT neurons that responded selectively to faces.



Question: Are there in IT, as for faces, selective cells for the different types of objects that can be encountered in the outside world (neurons for chairs, for flowers, for cars, etc ...)?

Neuron that responds to faces: The neuron responds to faces of different species (1, 4, 5). The discharge is reduced if the elements of the face are mixed (2) or occluded (3). The neuron does not respond to other biologically relevant stimuli (5).



Desimone, Albright, Gross and Bruce, J. Neurosci, 1984

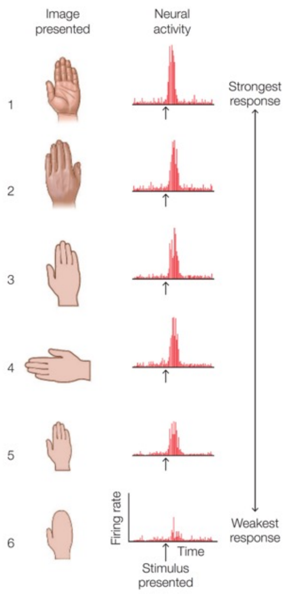
Recording of a single neuron from monkey IT cortex (Desimone et al., 1984)

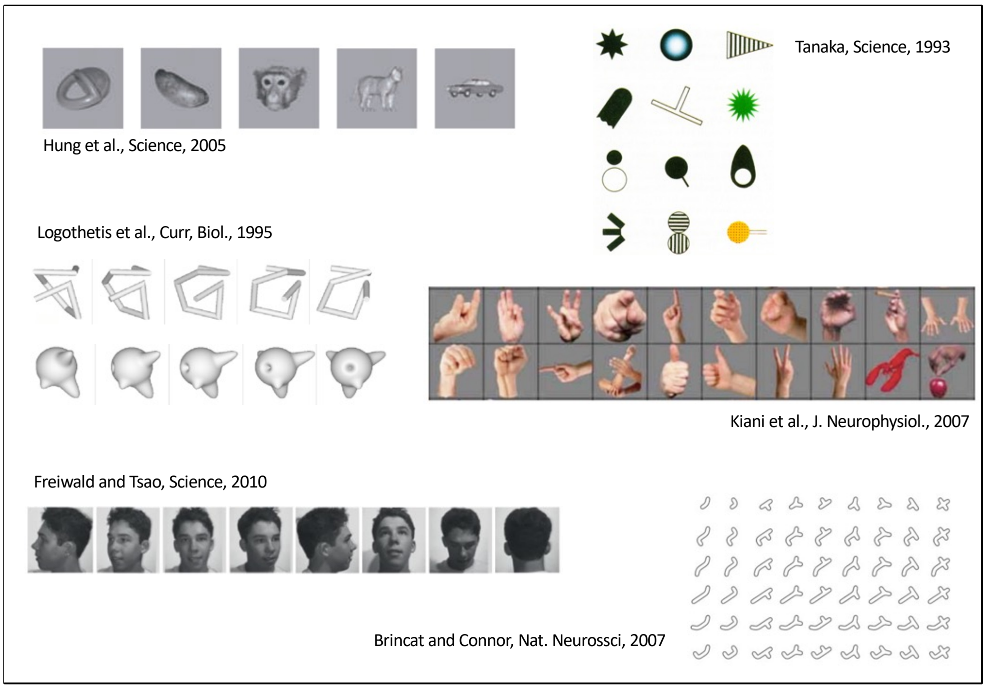
This cell is activated by the vision of the human hand.

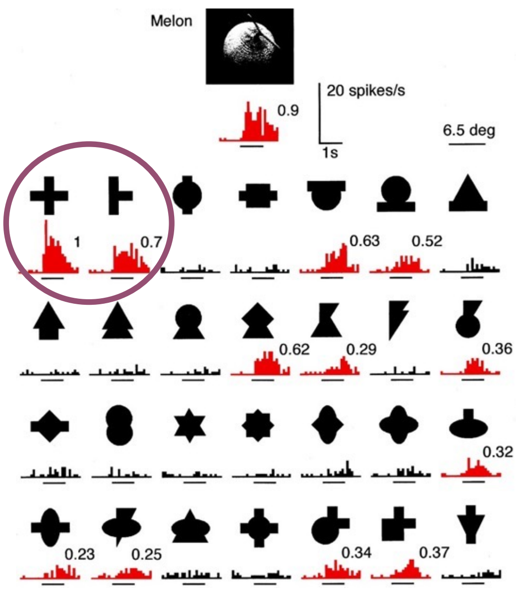
The first five images in the figure show the cell's response to various perspectives of a hand.

Activity is high regardless of hand orientation and only decreases slightly when the hand is noticeably smaller.

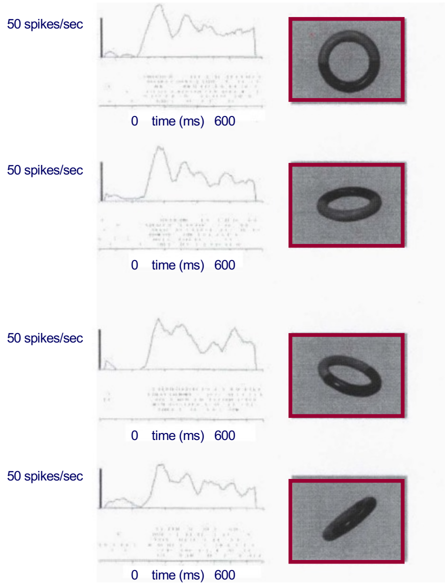
The sixth image shows that the response decreases if the stimulus has the same shape, but does not have well-defined fingers.




Tanaka, Science, 1993
Hung et al., Science, 2005
Logothetis et al., Curr, Biol., 1995
Kiani et al., J. Neurophysiol., 2007
Freiwald and Tsao, Science, 2010
Brincat and Connor, Nat. Neurossci, 2007



For example, the neuron shown in the figure seems to respond to the complex image of a fruit (melon). However, a detailed study (method of progressive reduction) shows that the neuron also responds to simpler stimuli that represent the visual elements of an object to which the neuron is sensitive.

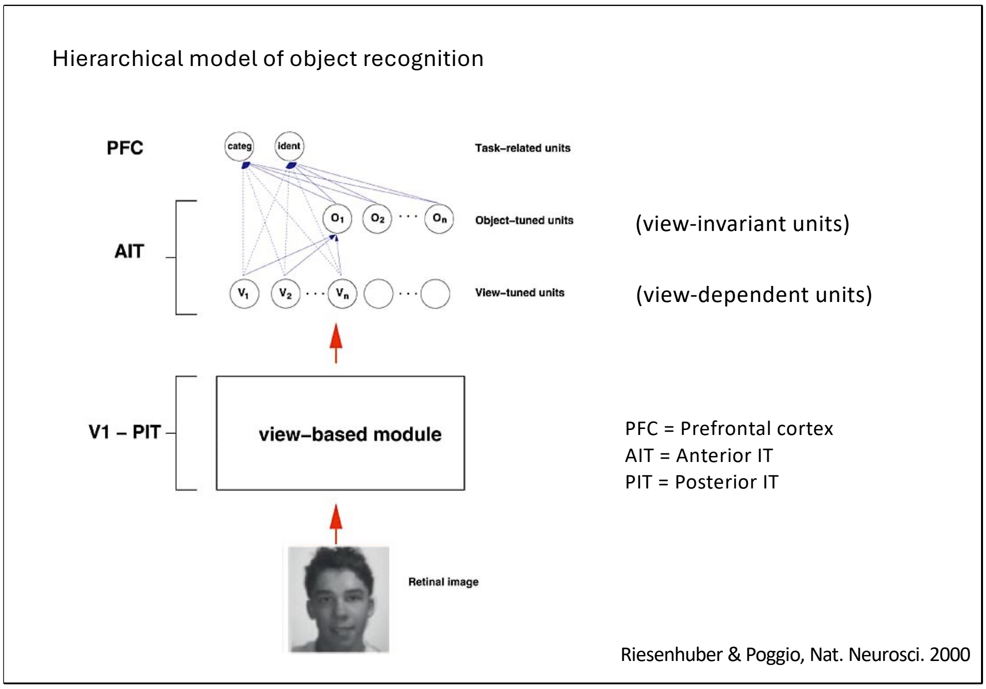


The majority of IT neurons respond to a stimulus only when it is presented from specific points of view (view-dependent responses).

Some neurons (10%) selectively respond to familiar stimuli regardless of their position with respect to the observer (view-independent responses).

These response, although rare, indicate that IT is capable of forming a (relatively abstract) representation of the object, rather than responding to one of the different forms that the object can take when its position with respect to the observer changes.

Logothetis & Sheinberf, Ann Rev Neurosci, 1996



HMAX Model Overview

The model is hierarchical and feedforward, inspired by the ventral visual stream:

S1: Simple cells (like V1) – oriented edge detectors.

C1: Complex cells – local invariance to position/scale via max pooling.

S2: Feature-combining units – tuned to intermediate-level features.

C2: Global invariance – combines S2 outputs to get invariant object representations.

View-dependent responses can emerge at intermediate layers (like S2), where units are tuned to specific views of an object.

View-invariant responses emerge at higher layers (like C2), due to pooling over multiple S2 units tuned to different views.

So, IT neurons can be both:

View-specific (important for fine discrimination).

View-invariant (important for robust object recognition).

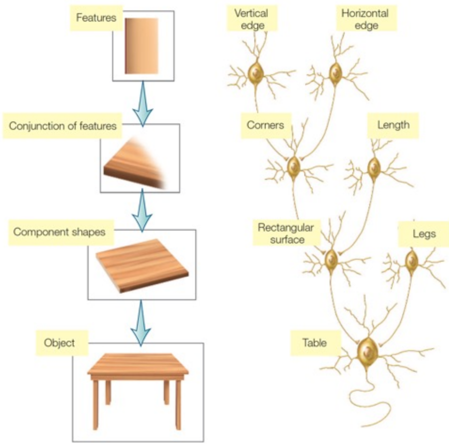
This hierarchical pooling reflects how the brain might balance specificity vs generalization.

The model bridged neuroscience and computer vision, influencing early deep learning ideas.

It was one of the first models to mechanistically account for real IT neuron behavior.

It showed how invariance could be built gradually from basic visual processing.

Hierarchical model of the object recognition

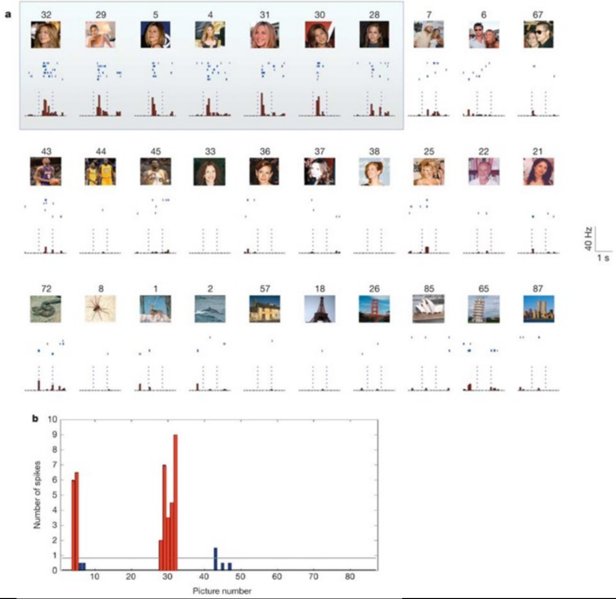


The finding that IT cells selectively respond to more complex stimuli than V1, V2 and V4 is consistent with a hierarchical model of object perception.

According to this model, each subsequent level encodes more complex combinations from the inputs of the previous level.

The type of neuron that can recognize a complex object has been called the gnostic unit, referring to the idea that the cell (or cells) signals the presence of
a complex, highly specific, and significant stimulus: that is, a known object, place or animal that has been encountered in the past.

Jennifer Aniston cell



Quiroga et al., Nature 2006

Local or distributed coding?

It is tempting to conclude that the cell represented by the activity of IT cells signal the presence of an object (a hand or face), independent of the point of view.

In this regard, the researchers coined the term 'grandmother cell' (Lettvin, 1969) to convey the idea that people's brains may have a gnostic unity that is activated only when the grandmother comes into view.

Other Gnostic units would specialize in recognizing, for example, a blue Volkswagen or the Golden Gate Bridge.

Jerome Lettvin at MIT, 1952

Distributed code hypothesis

An alternative to the Grandmother cell hypothesis is that object recognition is the result of a distributed activation pattern on the population of IT neurons.

According to this hypothesis, recognition is due not to one unit but to the collective activation of many units.

Distributed code theories easily explain why we can recognize similarities between objects (say, a tiger and a lion) and make mistakes between visually similar objects - both objects activate many of the same neurons.

Losing some units may degrade our ability to recognize an object, but the remaining units may be enough.

Distributed code theories also explain our ability to recognize new objects. New objects have a resemblance to familiar things, and our perceptions result from activating units that represent their characteristics.

The results of the studies on single neurons of the temporal lobe are in agreement with the theories of the distributed code of object recognition.
Although it is surprising that some cells are selective for complex objects, the selectivity is almost always relative, not absolute.

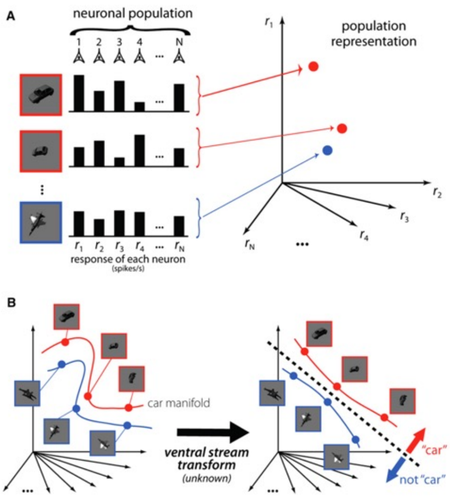
IT Cortex (Pre-ANN Era)

- IT neurons respond only to visual stimuli.
- The RF always include the fovea, that is the part of the retina most involved in the fine recognition of a visual stimulus.
- RFs tend to be large, providing the opportunity to generalize the stimulus within the receptive field, and often extend along the midline in both visual hemifields
- IT neurons encode complex characteristics of the stimulus (not simple features, such as color, form orientation, depth). A single neuron might respond strongly to a face, a hand, or a specific object category (e.g. tools, animals). Response is invariant to some changes in size, position, pose, and illumination.
- IT neurons often maintain selectivity even with partial or noisy stimuli. Suggests robustness and generalization, not pixel-perfect matching.

IT neuron selectivity often appears somewhat arbitrary.
A single IT neuron could, for example, respond vigorously to a crescent of a particular color and texture.
Cells with such selectivity likely provide inputs to higher-order neurons that respond to specific objects.

- IT neurons use a distributed representation: object identity is encoded across populations of neurons. Yet each neuron may have a sparse and relatively specific response profile—so the code is neither purely local nor fully distributed.
- Responses of IT neurons are shaped by experience and learning. Neurons can develop selectivity for new categories or become more finely tuned with exposure (e.g., training with novel shapes or objects). This supports category learning, memory, and conceptual abstraction.
- IT neurons fire ~100-150 ms after stimulus onset, showing that they're at the end of the ventral visual stream. This timing reflects a feedforward cascade through V1 → V2 → V4 → IT, with progressively more abstract representations.
- Some IT neurons correlate with object-related decisions, attention, or memory, especially in delayed match-to-sample tasks. This blurs the line between perception and cognition—IT is not just about visual features, but meaningful representations.

Ventral visual pathway gradually ''untangles'' information about object identity

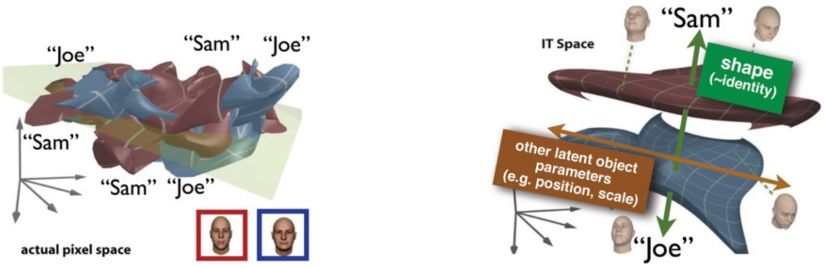


Response of a population of neurons to a particular view of one object can be represented by a **response vector** in a space whose dimensionality is defined by the number of neurons in the population.

When an object undergoes an identity-preserving transformation, it produces a different pattern of population activity, which corresponds to a different response vector.

Together, the response vectors corresponding to all possible identity preserving transformations define a low-dimensional surface in this high-dimensional space—an **object identity manifold.**

DiCarlo et al., Neuron, 2012



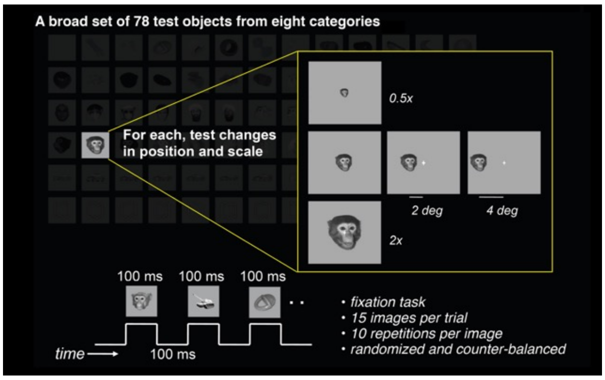
Object recognition is the ability to separate representation that contain one particular object from representation that do not.

Thus, object manifolds are thought to be gradually untangled through nonlinear selectivity and invariance computations applied at each stage of the ventral pathway.

At higher stages of visual processing, neurons tend to maintain their selectivity for objects across changes in view; this translates to manifolds that are more flat and separated (more ‘‘untangled’’).

DiCarlo et al., Neuron, 2012

For neurons with small receptive fields that are activated by simple light patterns, such as retinal ganglion cells and V1, each object manifold will be highly curved.
Moreover, the manifolds corresponding to different objects will be ‘‘tangled’’ together,



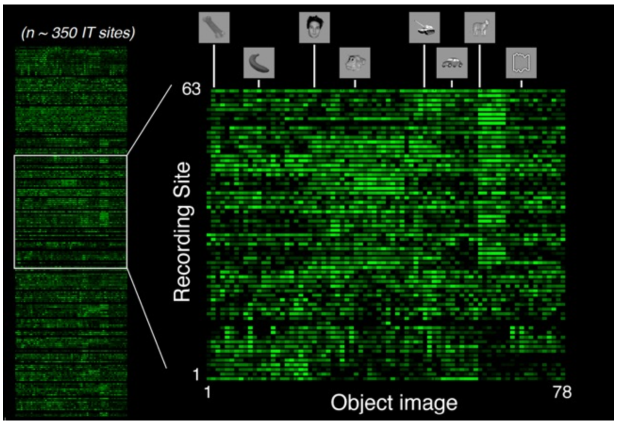
Categories: toys, food, human faces, monkey faces, hand/body, vehicles, white boxes, cats/dogs.

Readout of object Identity from IT cortex

By using a classifier-based readout technique, Hung et al (2005) investigated the neural coding of **selectivity and invariance** at the IT population level.

They showed that the activity of small neuronal populations (~ 300 units) over very short time intervals (as small as 12.5 milliseconds) contain accurate and robust information about both object ''identity'' and ''category.''

Hung et al., Science, 2005



The readout technique consists of training a regularization classifier to learn the map from neuronal responses (from the independently recorded neurons) to each object label.

The input consists of the neuronal responses of IT neurons

The activity of ~ 250 randomly selected multi- and single-unit activity in response to 78 images of different objects were recorded at 350 IT sites in two monkeys

Hung et al., Science, 2005

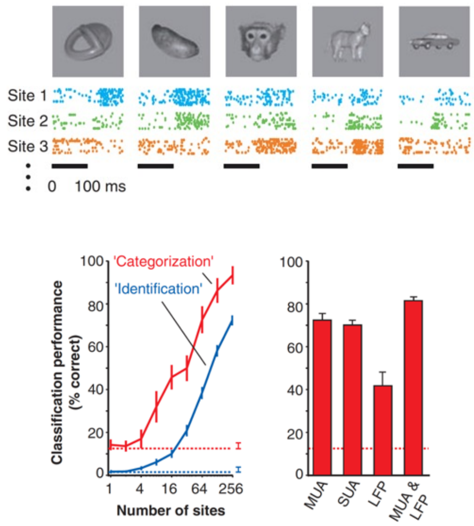
In Hung et al., 2005, during the training phase, the input to the SVM classifier was:

- A single feature vector per image,
- Where each element of the vector was the (average) neural firing rate of one IT neuron during a defined time window after stimulus onset (specifically around 100–300 ms after the image was shown).

They recorded from about 300 neurons (or "multiunit sites") in macaque inferior temporal (IT) cortex.

For each presented image, they built a vector with ~300 dimensions,

- Each dimension = firing rate of one neuron during the time window.

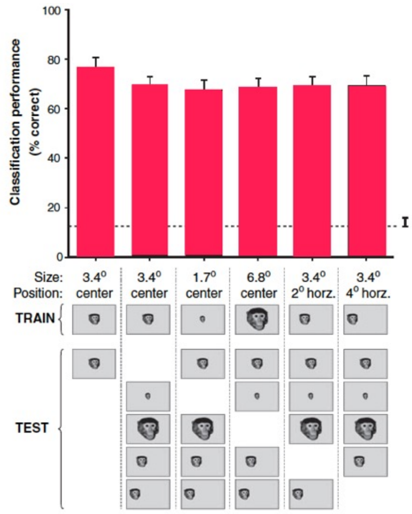


The spiking activity of 256 randomly selected multi-unit activity sites was sufficient to categorize the objects with 94% accuracy.

Classifier performance increased approximately linearly with the **logarithm of the number of sites**, which is indicative of a distributed representation in contrast to a grandmother-like representation.

Hung et al., Science, 2005

They trained the SVM classifier using neural responses to a subset of images, where each object appeared under various transformations.
Then, they tested the classifier on different images of the same objects — ones not seen during training — showing the objects in different sizes, positions, and backgrounds.



After training, the classifier was used to decode the responses to novel stimuli.

The classifier performance is maintained high over a range of object positions and scales, even for novel objects.

A one-versus-all approach was used whereby for each class of stimuli (8 classes for categorization, 78 classes for identification, 3 classes for scale and position readout), one binary classifier was trained.

Recognition performance is what downstream neurons (for instance PFC neurons) could, in theory, perform by simply computing a weighted sum of IT spikes over a short time interval (100- to 300-ms interval divided into bins of 50 ms in this case).

This is notable considering the high trial-to-trial variability of cortical neurons.

Hung et al., Science, 2005

Objects could be reliably categorized and identified (with less than 10% reduction in performance) even when transformed (spatially shifted or scaled),
although the classifier only saw each object at one particular scale and position during training.

Hung et a found that IT population contain also useful, although coarse, information about object scale and position.
There was little correlation between the ability of each IT site to signal scale/position versus object category information, suggesting that IT neurons encode both types of information

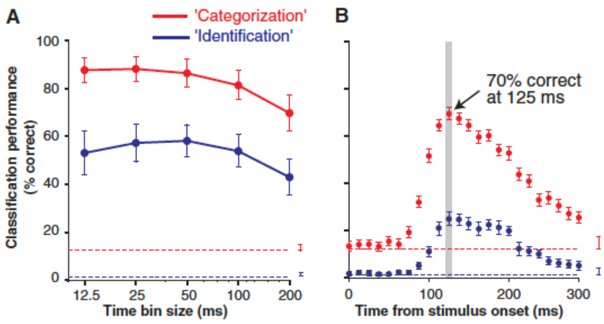
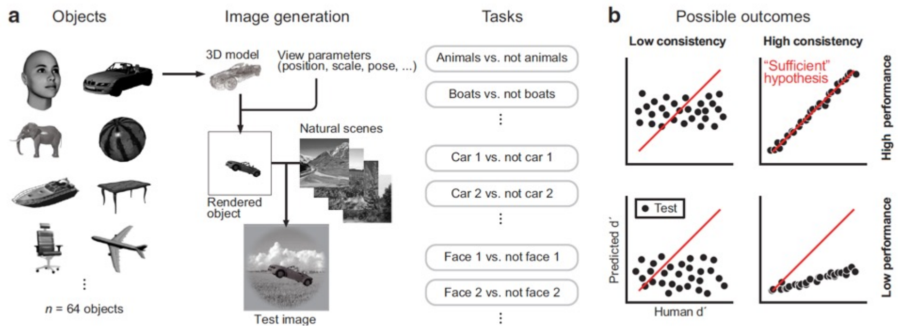


Fig. 3. Latency and time resolution of the neural code. (A) Classification performance (n = 128 sites) as a function of the bin size (12.5 to 200 ms, i.e., temporal resolution) to count spikes within the 100- to 300-ms window after stimulus onset for categorization (red) and identification (blue). The same linear classifier as in Figs. 1 and 2 was used. (B) Classification performance (n = 256 sites) using a single bin of 12.5 ms to train and test the classifier at different latencies from stimulus onset (x axis). The colors and conventions are as in Fig. 1B.

Hung et al., Science, 2005

Simple learned weighted sums of inferior temporal neuronal firing rates accurately predict human core object recognition performance



To determine whether models based on the firing rates of neurons in monkey V4 and IT cortex could predict human object recognition behavior.

They asked whether simple linear classifiers (weighted sums) trained on these neural responses can match human-level performance on object recognition tasks

Majaj et al., J. Neurosci, 2015

In a follow-up study, Majaj et al., (2016) investigated the relationship between neuronal activity in the V4 and IT cortex of monkeys and human object recognition performance.
In this research, the authors recorded neuronal responses from V4 and IT cortex of monkeys exposed to a diverse set of images representing various object categories with variations in position, size, and pose. Simultaneously, human participants were tested on their ability to recognize these objects under similar conditions.

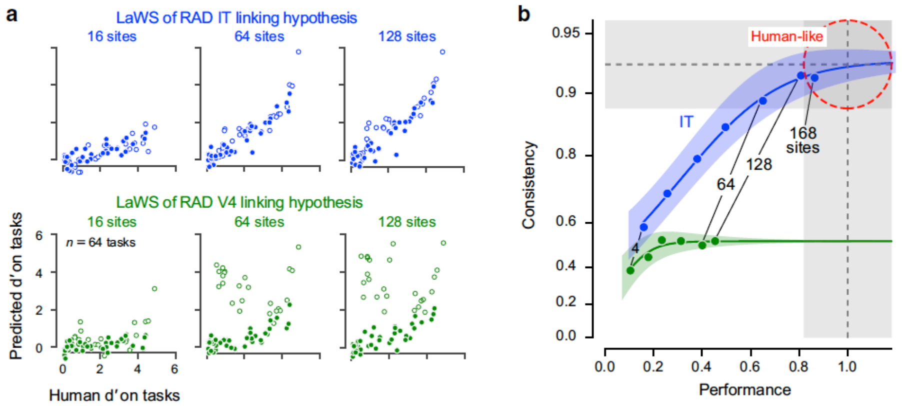
The key finding was that simple, learned weighted sums of the average firing rates from a distributed population of IT neurons could accurately predict human performance across 64 object recognition tasks.
By contrast, hypotheses based on low- and mid-level visually evoked activity (pixels, V1, and V4) were very poor predictors of the human behavioral pattern.

In Majaj et al., for each hypothesis, there were four possible outcomes, based on two factors:
1.Consistency: How well the model’s predicted d’ correlates with actual human d’ across 64 object recognition tasks.

1. High consistency: Strong correlation – the model predicts not just overall performance but which tasks are easier or harder, like humans.
2. Low consistency: Weak correlation – the model's pattern of success doesn’t align with human behavior.

2.Performance: How accurate the model is overall.

1. High performance: Model matches or exceeds human-level accuracy.
2. Low performance: Model performs worse than humans.



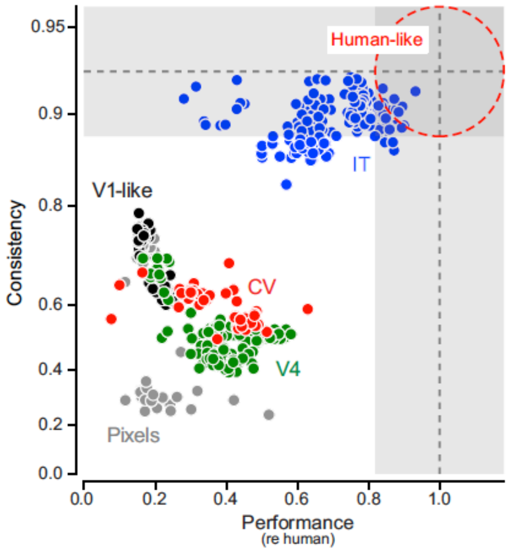
They measured how decoding accuracy improved as they added more IT neurons. As expected, more neurons = better prediction.

By contrast, adding more V4 neuron did not improve decoding accuracy.

LaWS of RAD = Linear Weighted Summation of Responses After Delay

Majaj et al., J. Neurosci, 2015

While they only recorded about 168 neurons, they extrapolated their results to estimate how well the brain could perform using all ~60,000 IT neurons.
This extrapolated "predicted d′" matched human performance, suggesting that the full IT population contains enough linearly separable information for high-level visual tasks, unlike V4 where adding more neurons gave diminishing returns.

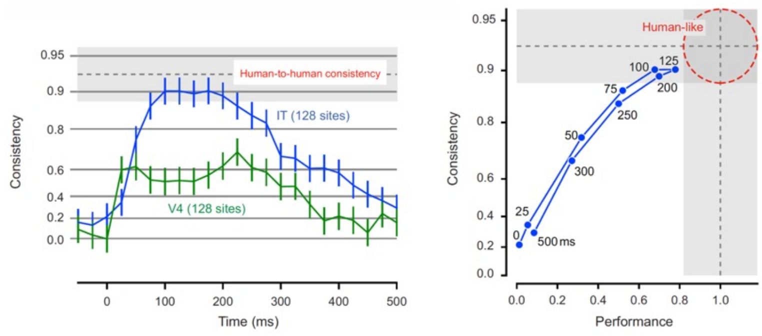


Majaj et al. tested 944 different models (linking hypotheses) to see how well neural data predicts human object recognition. Each model's **consistency** (match to human patterns) and **performance** (accuracy compared to humans) were plotted.

if we have the same accuracy we make the same mistakes

IT-based models (in blue) showed the best results, especially when more neurons were used, often coming close to or matching human performance. The ideal models both matched human accuracy and predicted which tasks were easier or harder for humans. This highlights that only models using rich IT data can fully explain human object recognition behavior.

Majaj et al., J. Neurosci, 2015



Majaj et al. showed that decoding IT activity depends on when you measure it. Using fixed 100-ms windows from 0 to 500 ms after image onset, they found that certain time windows yield higher consistency with human behavior. The best predictions came from neural activity shortly after image onset, revealing an optimal temporal window for matching human object recognition.

Majaj et al., J. Neurosci, 2015